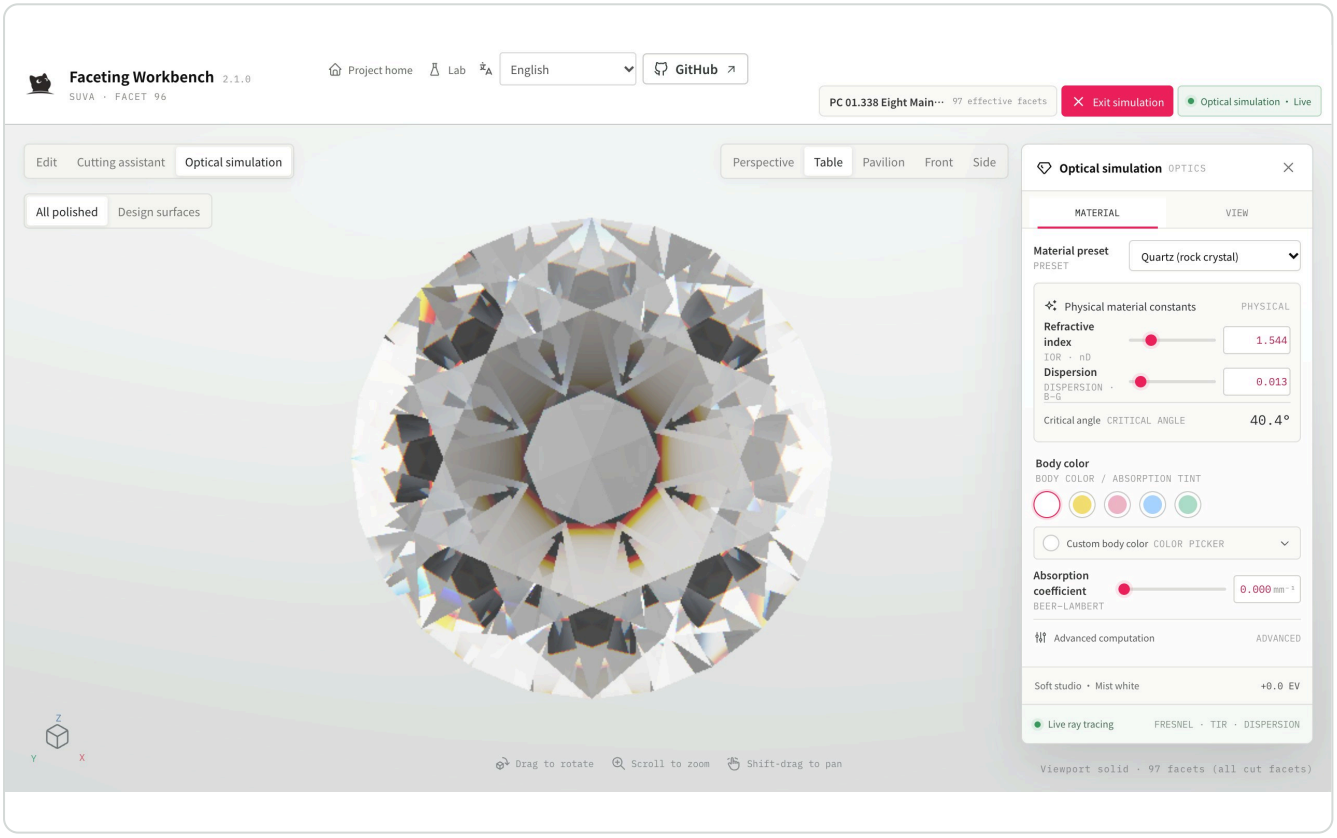
Designer’s illustrated manual

From an idea to a gem of your own

Start with outline and proportion, arrange facets into a rhythm, then compare materials and lighting. Give every action a visible design purpose.



Eight Main Highlight · Actual quartz-material view of preset PC 01.338; this is not a prediction of physical fire or a machining guarantee.

Who this manual is for

For designers who adjust cuts, compare alternatives and communicate with faceters. No engineering vocabulary is required: each example starts with the shape you want to achieve.

Getting started

Prepare the exercises in docs/manual/examples and open the version shown in the footer. Try composite tools on page 3, explore projects and starting points on pages 4–5, then begin the round-cut exercise on page 6.

Reading route

Find the design question you are working on

On your first visit, follow Start → Compare → Position → Deliver. Later, jump directly to the page that matches your design goal.

03–05 **Start here**
Composite tools, project home, new projects and starting points

06–09 **Choose a telling start**
Exercise files, preset cuts, shaped stock and your own rough

10–13 **Get to know the workbench**
Interface map, live previews, edit and save, angle, depth and direction

14–15 **Match the equipment and the proportions**
Green and red equipment lights, overall proportions

16–19 **Organize the facet rhythm**
Repeat, mirror and preform, custom primary, ring cut, arc cut

20–24 **Make facets meet where you want**
Jump, single point, edge ratio, dual Meet, stepping back

25–26 **Continue after changing earlier cuts**
Source repair, layer-by-layer trial and meetpoint inspection

27–28 **Concave cuts and cutting order**
Five-fold grooves, replaying the cuts in real order

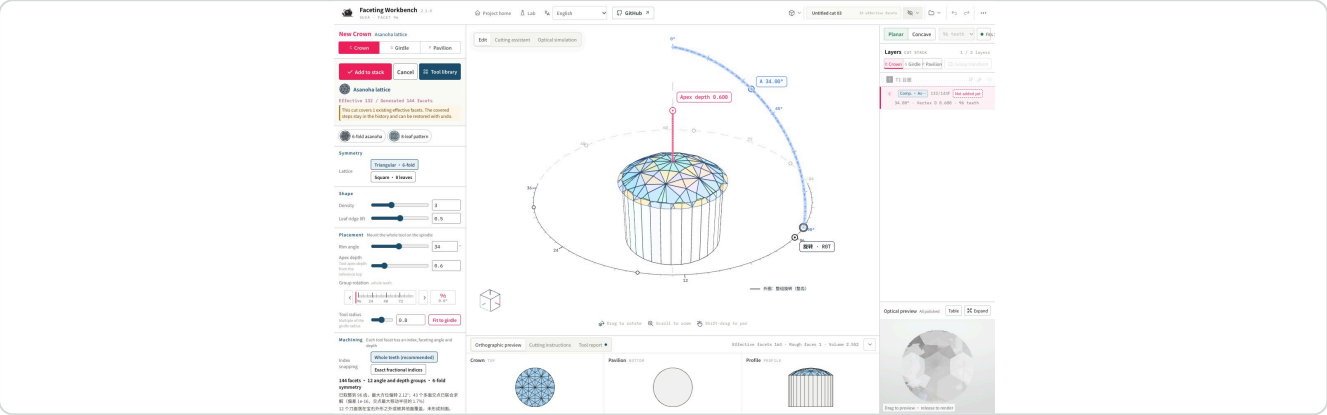
29–31 **Compare materials, bring back experiments**
Optical simulation, design review, pattern experiments and resuming older drafts

32–40 **Deliver, exchange and collaborate**
Files and format center, project saving, reference trials, conversation hand-off, delivery checklist

Tool library · a shape exercise

Shape a facet group with one composite tool

Ring cuts, grids, tier assemblies and fancy studies share one library. Choose a region and tool, adjust shape and placement, then add the whole tool as one layer.



Default 96-tooth start, crown asanoha: triangular six-fold, density 3, rim angle 34°, apex depth 0.600, tool radius 0.8, whole-tooth indexing.

1. **Choose a region and tool**
   Create a default 96-tooth project. Choose Crown on the left, start a new cut layer and select Asanoha lattice in the tool library.

2. **Adjust and compare**
   Set triangular six-fold, density 3, rim angle 34° and tool radius 0.8. Change apex depth from 0.550 to 0.600. Keep the same top and side views to compare leaf edges and crown height.

3. **Review the tradeoff before saving**
   Open Tool report below the canvas to inspect effective facets, meets and short-edge warnings. Choose Add to stack at the upper left. The tool is one layer and one undo step.

4. **Continue editing a saved tool**
   Click the layer on the right to adjust the same tool. Older tools retain their saved algorithm. To edit individual levels, leave editing and choose Dissolve on the layer.

Pause and make a design judgment

Under the same viewing conditions, are the leaf edges clear and the crown height suitable? If short edges appear, compare exact fractional indexing with whole teeth.

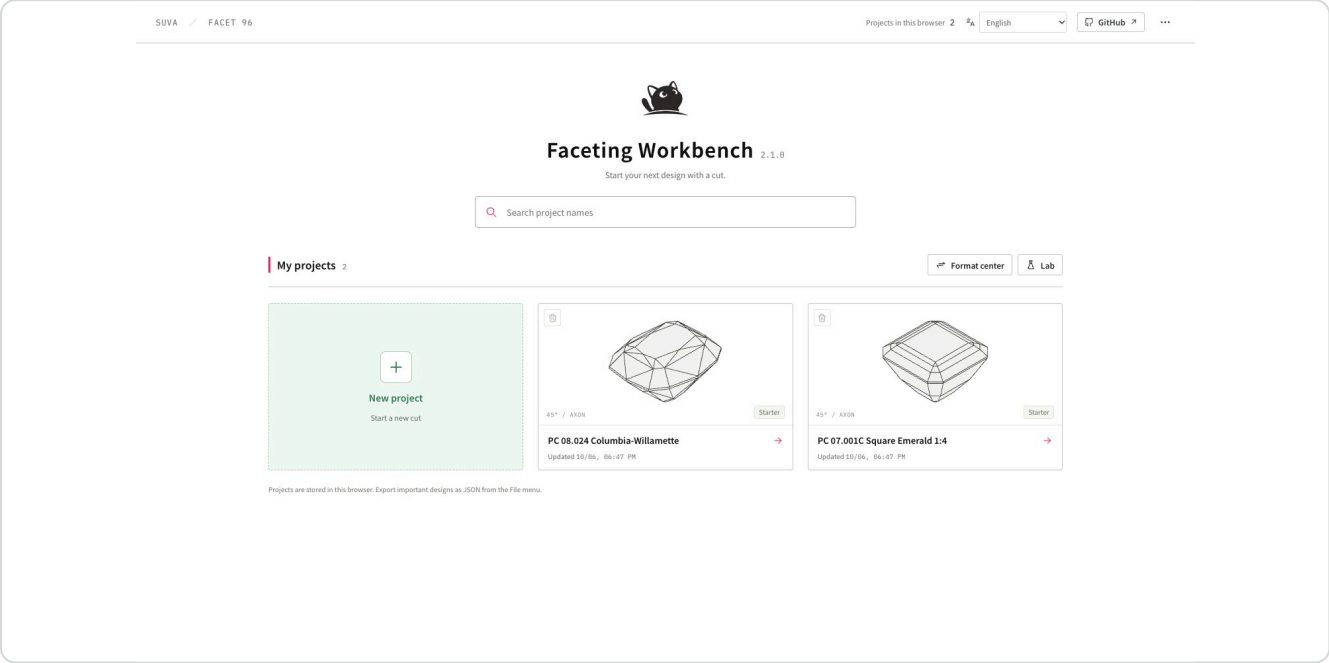
Before you continue

Tool reports describe simulated geometry and indexing compatibility, not guaranteed cutting or optical results. Ring tools support Meet/Jump on the primary facet; dissolve grids and other composite tools before positioning individual levels.

START · PROJECT HOME

Find the project you want to continue

The workbench opens at project home. Treat each project as an evolving design: name its purpose and use the actual outline on its card to recognize it.



Your first visit includes two real cuts. New project and saved projects share a left-to-right grid, with each new row starting at the left edge; All expands the rest, and search shows every match.

1. **Create a starting point**
   Click + New project, choose the equipment wheel, then a default start, a preset cut, stock presets or an uploaded blank; the next page explains the starts.

2. **Continue a design**
   Search by name and open its card. Reopening a project saves to the same card, without making a copy. Rename the design at the top of the editor.

3. **Keep your current experiment**
   Returning home leaves the current CUT intact. “Resume design” restores that workspace; switching or creating a project asks about any unsaved preview.

Pause and make a design judgment

Can you recognize the design by its name and outline? If not, give it a short name that expresses the design goal.

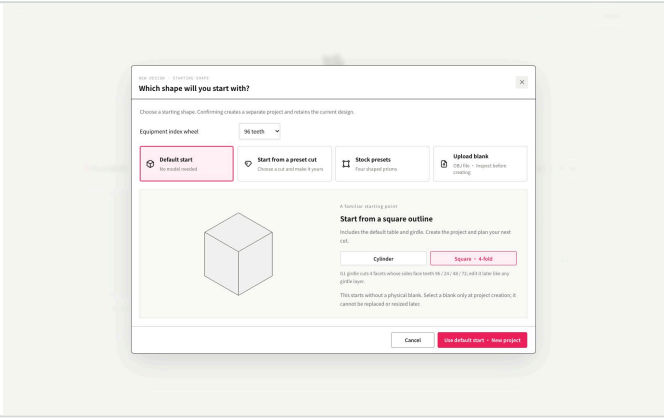
Before you continue

Committed cuts and material save automatically. Cards exclude unsaved previews; reloading does not restore drafts or old undo history. The card’s upper-left delete control requires confirmation and cannot be undone. Export important designs as JSON.

Getting started · New project

New project: equipment first, then the start

Whether a gem cuts on whole teeth on your machine is decided when you create it. Choose the equipment wheel first, then the start: default start, preset cut, stock presets or uploaded blank.



96-tooth default start with Square · 4-fold: the preview becomes a square prism at once, with the G1 girdle sides facing teeth 96 / 24 / 48 / 72.

1. **Choose the equipment wheel**
   Click + New project and pick your machine's tooth count under Equipment index wheel. All planar cuts and the girdle then use this wheel; you can still change wheels in the project without changing the shape.

2. **Choose the default start outline**
   The default start has the fixed table T1 and girdle G1. With Cylinder, pick a count under Girdle facets: only counts of 8 or more that the wheel divides on whole teeth are listed, defaulting to the one nearest 32. Square · 4-fold gives a square prism with sides on whole teeth; wheels not divisible by 4 (77, 99 teeth) cannot choose it, and the dialog says why.

3. **Compare, then create**
   The preview on the left follows each outline or count change. Click Use default start · New project when ready; G1 remains an ordinary girdle layer you can edit later or turn into a ring or arc cut.

4. **Other starts**
   Start from a preset cut loads an existing design; Stock presets and Upload blank start from a shaped outline or your own rough with no cuts. See pages 7–9.

Pause and make a design judgment

Does your design start better from a round or a square outline? Does the girdle count suit the crown repeat you plan next?

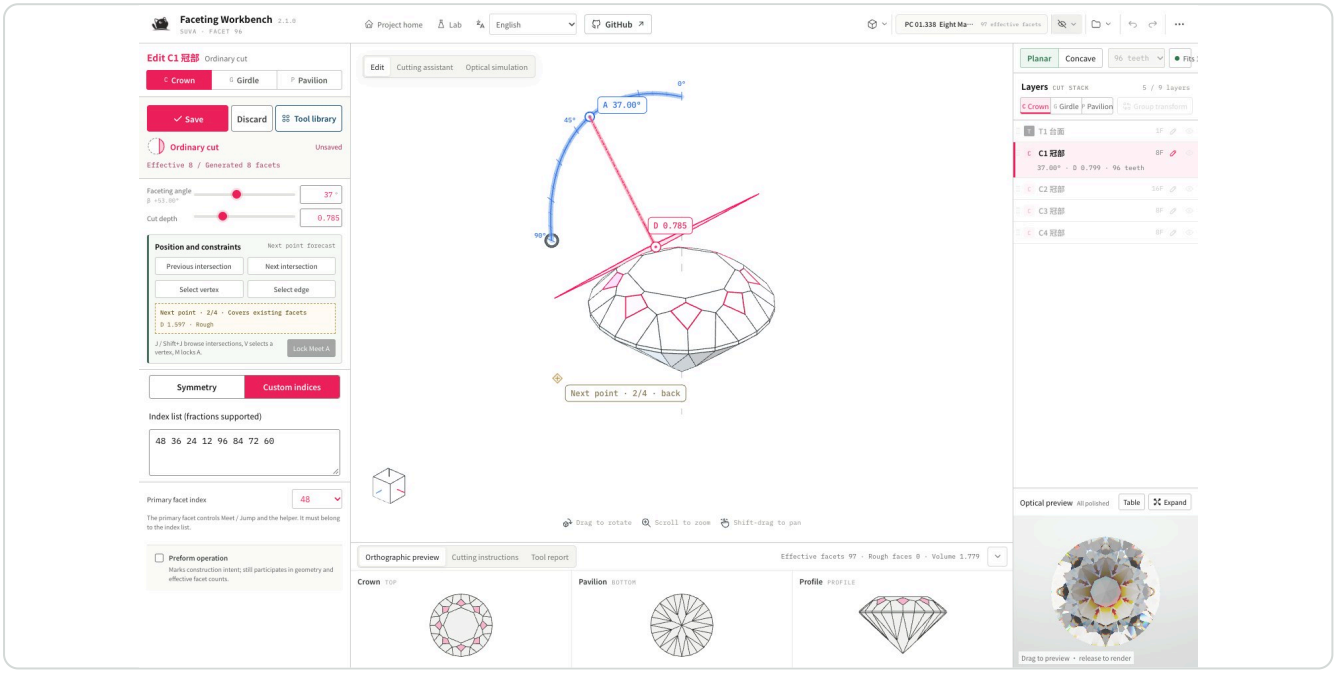
Before you continue

Stock is chosen only when you create a project and cannot be replaced or scaled later; the index wheel can be changed in the project.

PREPARE · EXERCISE FILES

Choose a starting point that reveals the question

Use Eight Main Highlight from the preset library for finished-cut observation. Cutting and Meet exercises use a simplified round built in stages, making new intersections easy to see. Exercises are not recommended finished cuts or production recipes.



Finished example: PC 01.338 Eight Main Highlight, 97 effective facets, designed by Long, R H & Steele, N W. On the home page choose New project → Start from a preset cut and search for its name.

Open the exercise

Enter the workbench, then choose File → Import JSON and select 01-round-start.json. It opens as a separate project, retaining the original. An unsaved preview requires confirmation before switching. Rename it, for example Round_Facet_Study_01, and export JSON.

Write one design goal

For example: “Keep the round outline and add a light decorative rhythm to the crown.” Judge each later change against this sentence.

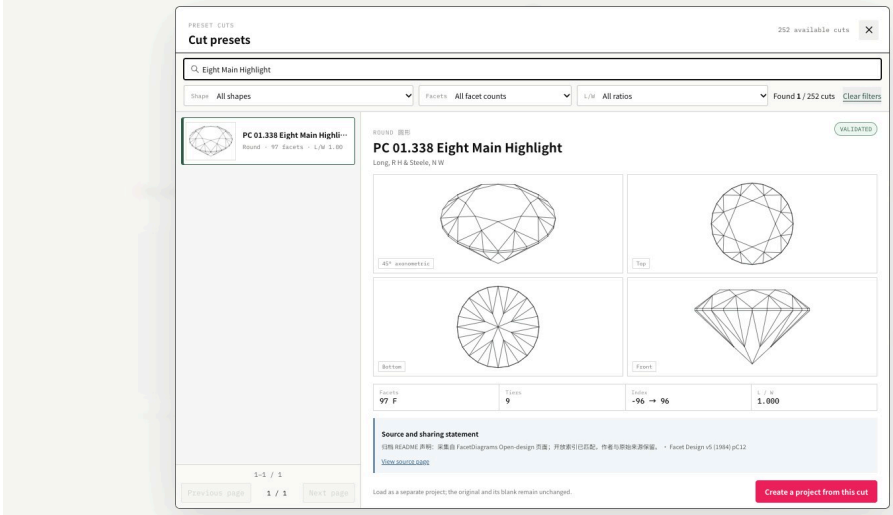
Exercise and comparison files

02 / 03 compare two ratios on the same edge; 04 uses two points; 05 lowers the crown; 06 adds four-direction decoration; 07 needs repair. File numbers identify alternatives, not a required cutting sequence.

START · CHOOSE A CUT

Start with an outline in the preset library

If you already want a cushion, oval or emerald shape, find a similar outline in the library. Treat the preset as a study: retain its author and source, and save your own variation separately.



Actual preset library: compare all four views to avoid perspective distortion.

1 **Open candidates**

Home → New project → Start from a preset cut. Combine outline, effective facet count and L/W filters; search names, authors or sources with multiple keywords. Each page shows 60 cuts.

2 **Look from four directions**

Top shows outline and table; front shows crown height and pavilion depth; bottom shows pavilion structure; axonometric shows the whole shape.

3 **Load and name**

Choose Create project from this cut, then rename your design and export a JSON copy as the starting version. Your original project stays on the home page. Filters and pagination do not change it.

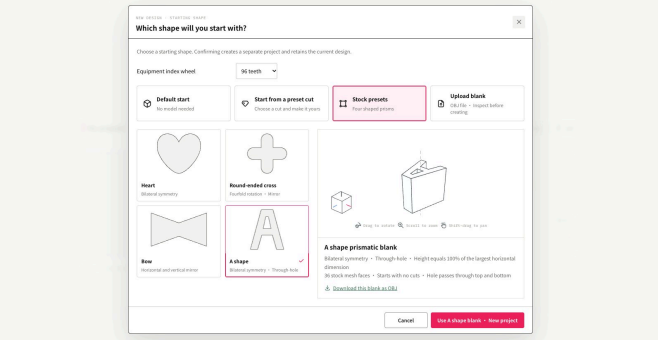
Pause and make a design judgment

Does the outline suit the intended setting? Does the table size suit the shape’s character? Describe any remaining concern as one concrete change.

STOCK PRESETS · ORIGINAL OUTLINES

Design from a shaped blank

Start directly with a heart, round-ended cross, bow or an A shape with a through-hole, without preparing a model.



Actual starting-shape chooser. Cross-section cards identify the outline; rotate the solid preview to inspect thickness and holes.

1. **Choose a design starting point**
   New project → Stock presets → A shape. All four blanks have height equal to 100% of the largest horizontal dimension. A has bilateral symmetry and a through-hole; heart is bilateral, cross has fourfold rotational and mirror symmetry, bow mirrors horizontally and vertically.

2. **Inspect before creating**
   Inspect the solid preview and choose Use A-shaped stock · Create project. With zero CUTs, start a crown cut on the left. Cancel or Back leaves the original design intact.

3. **Compare before and after cutting**
   Try one cut and adjust depth while observing the A's hole and legs. Save and undo to compare outline retention at the same angle. Symmetric rough does not automatically mirror cutting instructions.

4. **Test your import workflow**
   Download the selected blank as OBJ and reimport through Upload blank. Keep units unspecified and +Z up; check the 100% height and through-hole. The default starting option still includes its table and girdle.

Pause and make a design judgment

Do the preset's recesses, holes and thickness suit your starting point in top and side views? Does one cut remove only what you intended?

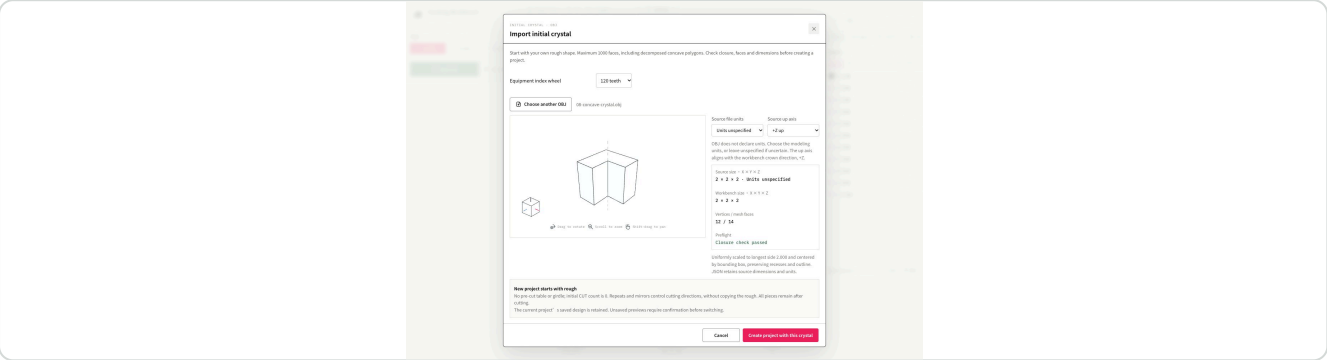
Before you continue

OBJ downloads in stock-presets/ use the same outlines as the built-in presets. A blank is a starting shape, not a finished gemstone cut.

YOUR CRYSTAL · OBJ IMPORT

Keep the rough’s recesses, then plan the cuts

For a starting shape with recesses or an irregular outline, preserve its shape before deciding what to remove. Repeats and mirrors arrange cutting directions; they do not make asymmetric rough symmetric.



Actual preflight of exercise 08’s L-shaped crystal: inspect the recess, source dimensions and up axis before creating a separate project.

1. **Import a real starting shape**
   New project → Upload blank, or File → Import initial crystal. The 1,000-face limit includes decomposed faces; failed preflight explains why. Open docs/manual/examples/08-concave-crystal.obj: an original L-shaped exercise measuring 2 × 2 × 2 in unspecified units, +Z up.

2. **Inspect shape and proportion**
   Keep Units unspecified and +Z up. Rotate the preflight model and check the L-shaped notch. For your own OBJ, choose its original units and orientation. The longest side scales uniformly to 2.000 and the bounding box is centered.

3. **Start your cutting sequence**
   Choose Create project from this crystal. There are no CUTs, automatic table or girdle. Start a cut from the upper-left command area, use one repeat first, and compare the original recess with the new facet.

4. **Keep a recoverable design**
   All pieces remain if a cut separates the solid. Clear cuts restores the original rough. JSON saves the rough and operations; crystal projects do not support ASC export, as preflight explains. Canceling import preserves the old project; switching asks about unsaved previews.

Pause and make a design judgment

Compare before and after in the same top and front views: did the new plane preserve the recess you wanted? Which repeated directions actually intersected the rough?

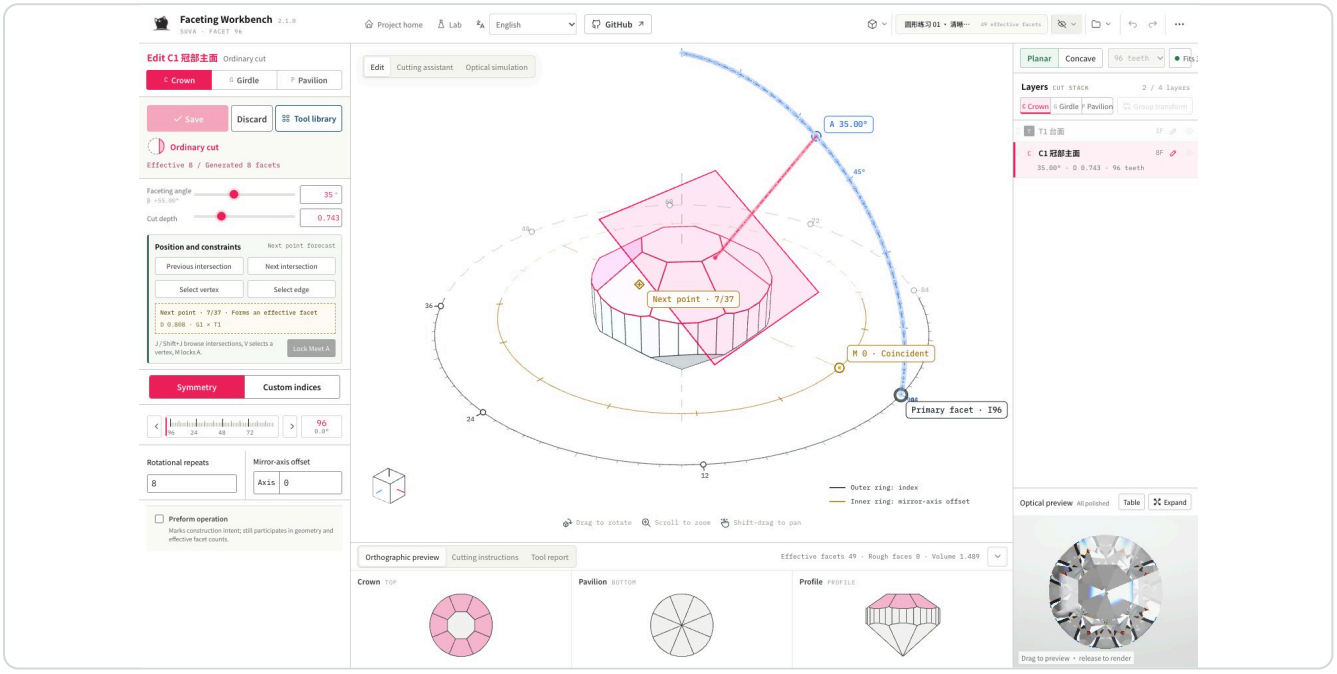
Before you continue

This exercise teaches the relationship between rough and cutting direction, not crystal measurement or a production recipe. If preflight fails, repair closure and face connectivity in a modeler, then re-export OBJ.

START · WORKSPACE MAP

Find your next action in the workbench

The center shows the 3D shape. The left operation panel holds region, tool, parameters and saving; the right layer panel manages saved operations. Below the canvas are orthographic views, instructions and tool reports.



Click the C1 layer on the right to edit it. Parameters appear on the left, with Discard and Save in its upper command area.

Left: adjust the current facets

Angle controls slope, depth controls advance and index controls direction. Ordinary cuts use repeat and mirror; composite tools use shape parameters. The upper-left command area always holds creation, cancellation or commit actions.

Right: manage layers

Filter layers by C Crown, G Girdle or P Pavilion. Click a row to edit, double-click its name to rename, or hover while idle for Dissolve and Delete. Creation and saving are in the upper-left command area.

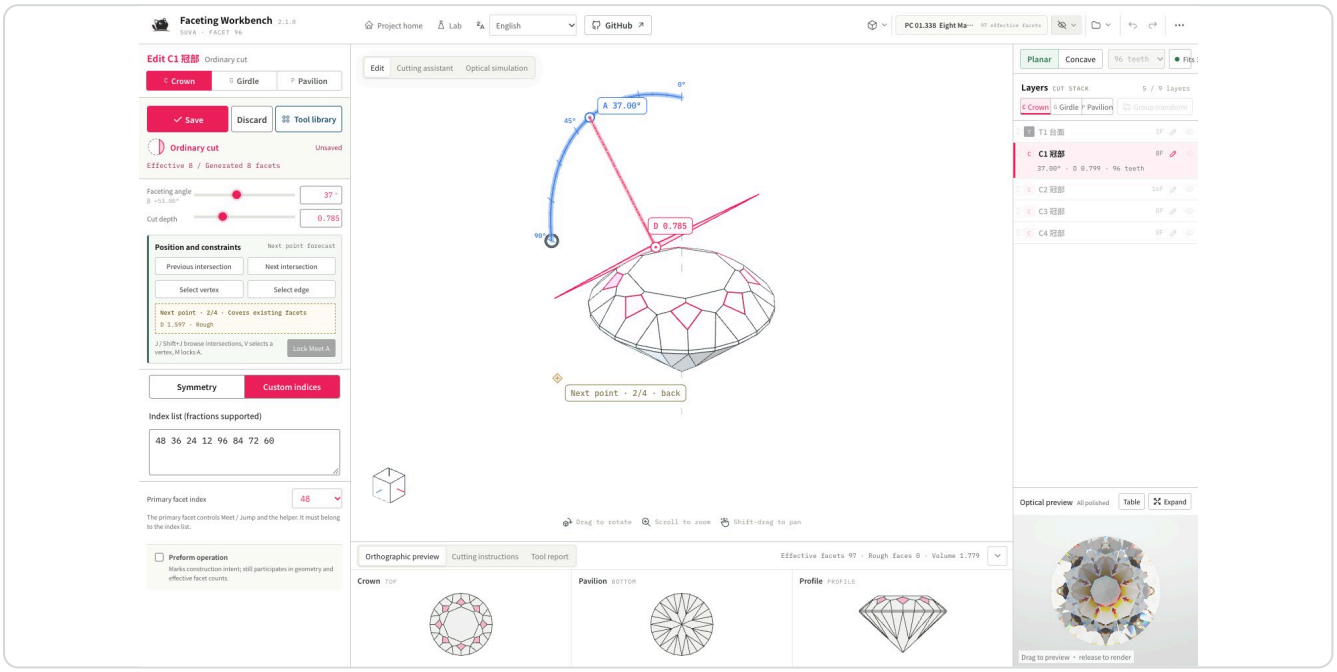
Top and canvas: inspect and manage

The top bar provides Home, Labs, view, display, file and undo controls. Switch between editing, Cutting Assistant and optical simulation at the upper left of the canvas. Drag to rotate, scroll to zoom, and Shift + drag to pan. Technical Support remains available on every page and in optical simulation, with the official team, testing group, GitHub and feedback links. External links open in a new tab; close the panel to resume your design.

Observe · orthographic views below the canvas

Check crown, pavilion and profile as you work

Judge facet rhythm and overall height together. Orthographic views below the canvas show crown top, pavilion bottom and side views of the current shape.



Crown, pavilion and side views appear side by side without the manipulator, keeping facet edges clear.

1. **Fix a starting point**
   Load Eight Main Highlight, click its C1 crown layer and slightly adjust depth. Watch the main view and all three lower views update together.

2. **Compare facet groups**
   Use the lower crown view to inspect facets around the table and the pavilion view to inspect bottom edges. These three views are read-only.

3. **Compare proportions**
   Check crown height, girdle and pavilion depth in the side view. Review the table position, outline transitions and small facets; save or discard in the upper-left command area.

> **Pause and make a design judgment**
> At the same viewing angle, does the added facet group make the rhythm clearer? Does the profile still match your design goal?

> **Before you continue**
> Side windows are live, read-only views: pale blue marks a new preview and pale pink the edited tier. Unsaved shapes can be compared but are excluded from projects and exports.

BASICS · EDIT AND SAVE

Edit a facet group by clicking its layer

Click a layer row or pencil on the right to edit it. The selected row shows a parameter summary, with full controls on the left. Double-click the layer name to rename it.



Tier list · Editing C1



Operation panel · discard or save

C1 is being edited; Discard and Save are in the upper-left command area.

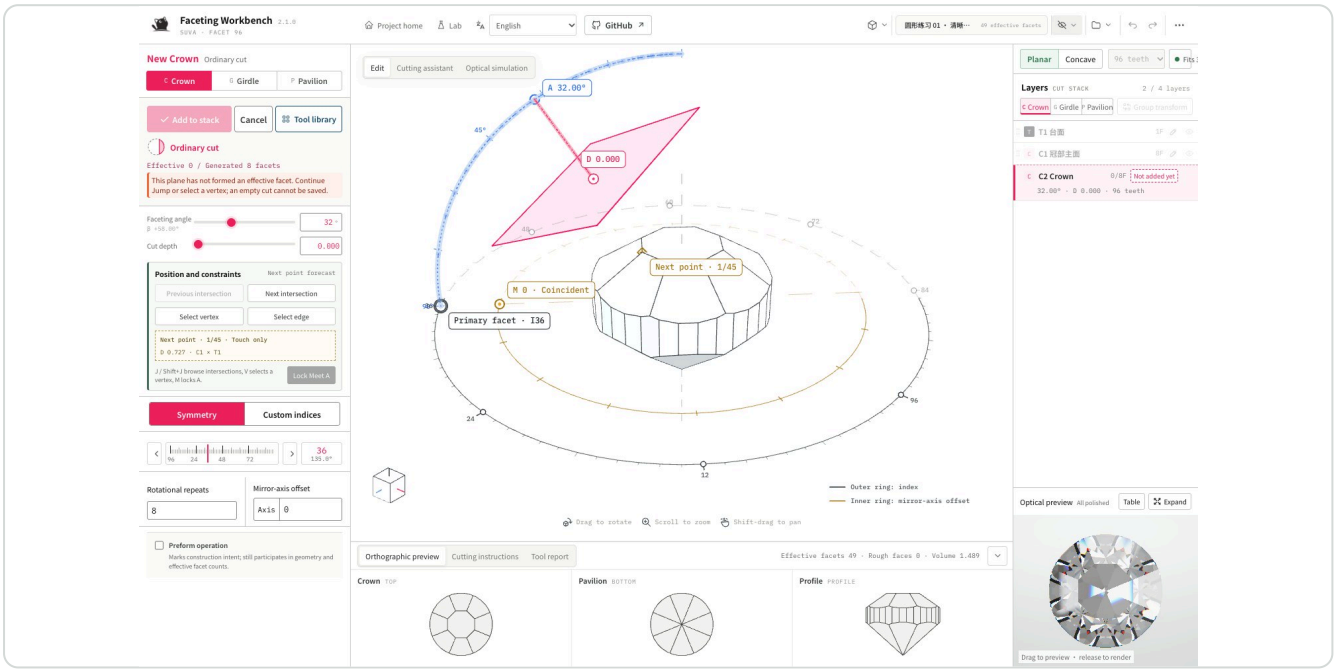
1. **Enter**
   Find C1 crown mains in the Crown list and click the row or pencil. Initially the stone keeps its saved shape.
2. **Try**
   Increase the angle slightly and use front view to inspect the tilt. Pink indicates the current edit, not the final material.
3. **Decide**
   Save the change if it works, or discard to retain the original. Saving returns to idle; Undo restores the previous step.

> **Pause and make a design judgment**
> Can you immediately tell which facets you are editing? Does the change support your stated goal? Being able to click Save is not the design outcome.

BASICS · WHAT THE PARAMETERS DO

Explore angle, depth and direction in small steps

Change one relationship at a time so you can identify its effect. Look at geometry before material. Blue new-cut previews and pink edit highlights are interaction feedback only.



The helper and primary plane show the direction of a new CUT. Actual facets are determined by intersections with the solid.

Angle: change the tilt

Crown and pavilion angles adjust in 0.01° steps. The table stays at 0°, the girdle at 90°. Single Meet allows angle changes; dual Meet determines the angle from two points.

Depth: decide how far to advance

Advancing the cutting plane along its normal changes intersections with neighboring facets. Enter exact values when needed. The handle endpoint is not a gemstone vertex.

Index: turn the facet group

The default wheel has 96 teeth. Choose another in Indexing and compatibility; the last tooth and 0 share a direction. Click the readout to enter fractions; dragging snaps to whole teeth. The gizmo follows the selected primary facet.

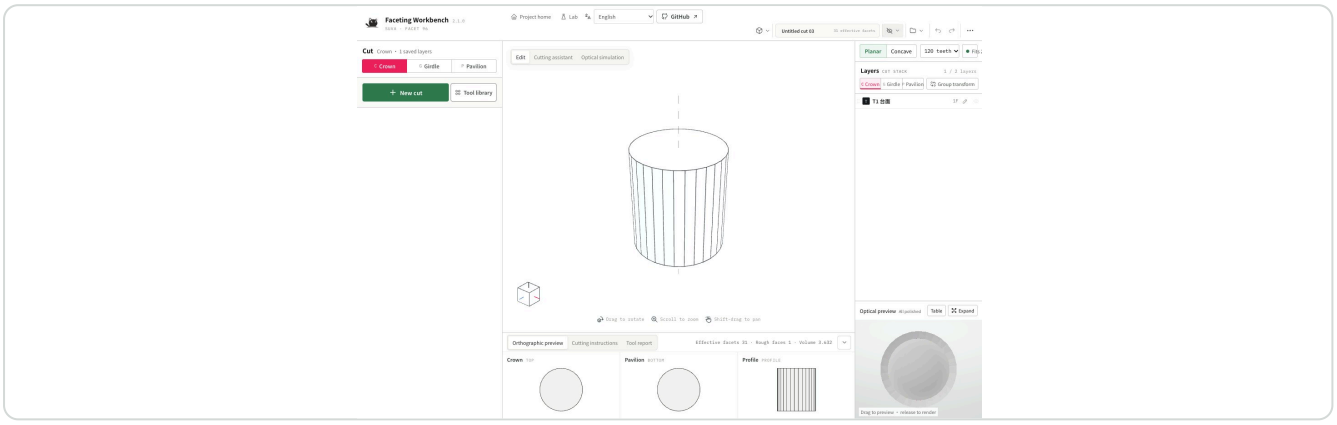
Why saving may be unavailable

A new cut needs a valid geometric result. It may completely cover an earlier facet; the earlier operation remains recoverable. Adjust or cancel if the cut removes all stock, makes no effective cut, or cannot satisfy its constraints.

Equipment wheel · Green and red

Choose the equipment wheel before designing

Choose the wheel supported by your equipment at project creation. Planar cuts and the girdle share it; stock and concave cutters remain independent shapes.



The default 120-wheel project has a 30-fold girdle. Compact green and red dots show compatibility.

1. **Start with a defined project**
   Home → New project. Choose a 120 wheel and use the default starting shape. Keep the perspective and side views unchanged.

2. **Check your wheel**
   The dot beside 120 is green. Girdle instructions use 120 teeth, without a 32-fold girdle inherited from a 96 wheel.

3. **Compare wheel changes**
   Switch to 96. The shape stays intact, all planar readings convert to 96 teeth, and incompatible directions produce a red dot.

4. **Return to your equipment**
   Undo the wheel change or select 120 again. Existing fractional directions are never silently rounded.

Pause and make a design judgment

Does the outline stay the same after switching wheels? Can you see compatibility at a glance?

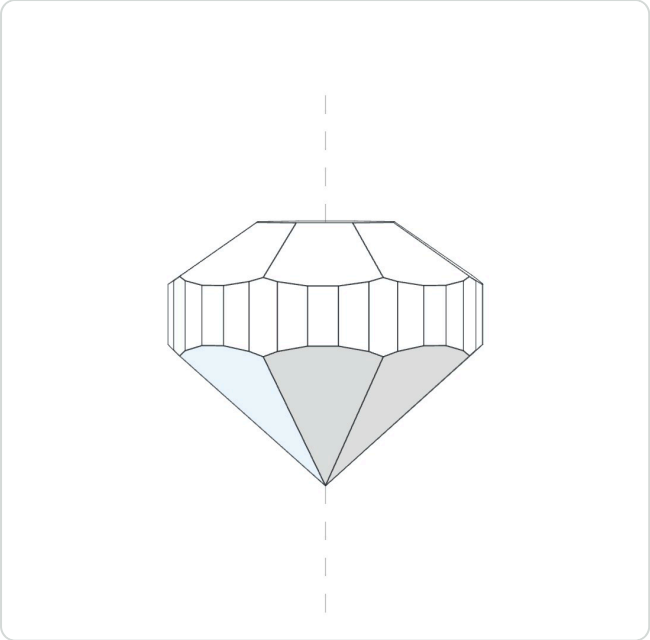
Before you continue

Dots check planar directions only, excluding stock and concave tools. Exercise 09 provides an additional legacy design for exact conversion checks.

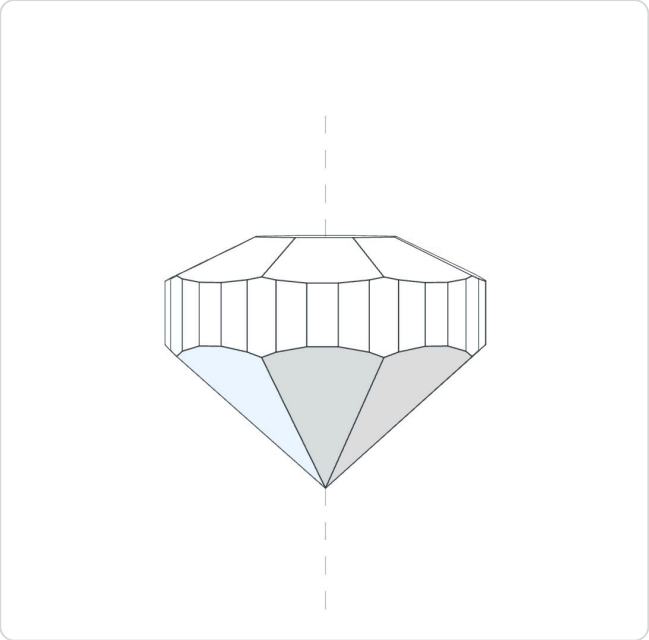
EXAMPLE A · OVERALL PROPORTIONS

Compare a tall crown with a lower crown

Goal: keep the round outline and see whether a lower crown suits your setting height or style. The group transform includes the table, helping retain continuity between tiers.



Original · Height 100%



Lower crown · Height 70%

Compare from the same front view. Only crown height changes; the pavilion stays unchanged.

1. **Open the original**
   Import 01-round-start.json and click Group transform in the crown list.
2. **Try 70%**
   Set height scale to 70%, leaving translation and rotation at 0. Inspect front and side views, then save. Import 05-low-crown.json to compare the result.
3. **Judge, rather than copy**
   Check the table-to-crown transition and whether the height suits your design. 70% is a comparison condition, not a universal optimum.

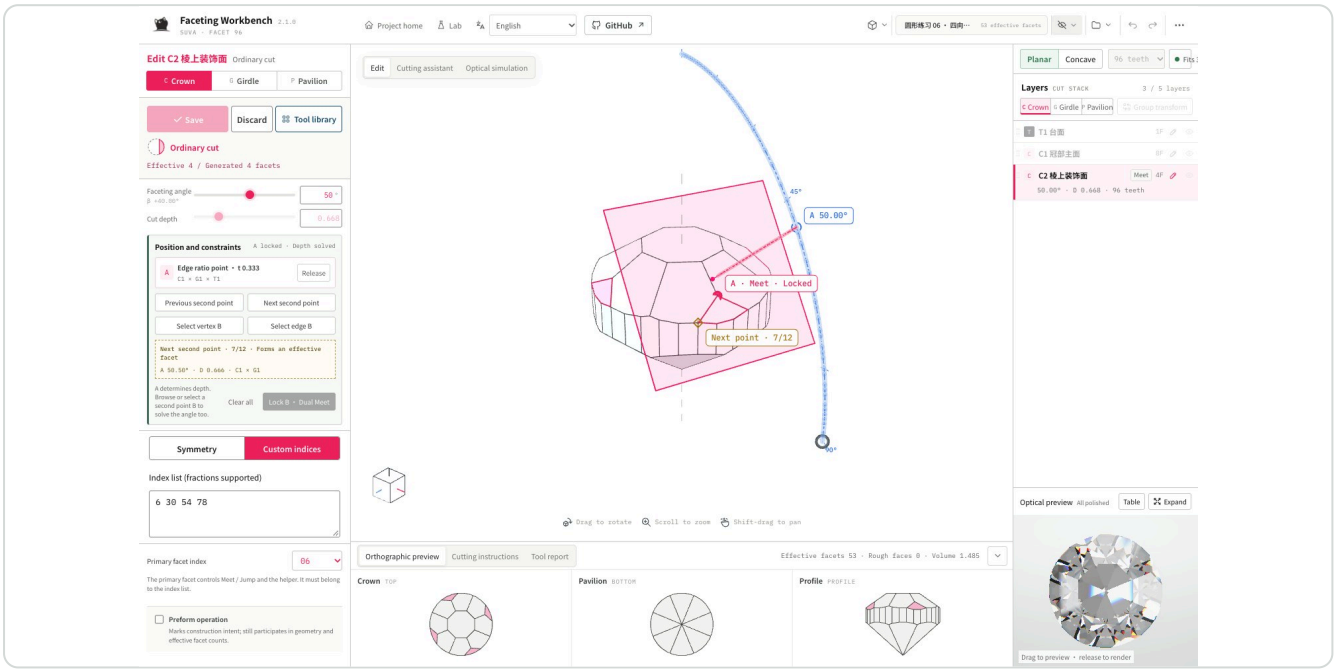
Before you continue

Group transforms combine translation and index rotation. The crown includes the table; the pavilion is separate. Height follows the planar outline; concave tools and the blank stay fixed, and the viewport shows the combined result. Existing Meets may need review after a source tier changes.

ORGANIZE · BEYOND FACET COUNT

Express intent with repeats, mirrors and preforms

A facet group may create rhythm or provide a source for a later cut. Meaningful step names make it easier to resume than remembering long parameter lists.



The helper and Meet / Jump follow the selected primary facet.

Repeat: one rule around the stone

8 repeats generate eight explicit planes. Later cuts may cover some earlier facets; the original operations remain available for undo and redo.

Mirror: add a second set of directions

Adjusting mirror-axis offset adds mirrored members; coincident planes are deduplicated. Observe the alternating rhythm in top view instead of maximizing facet count.

Preform: describe this step’s purpose

Mark a normal crown or pavilion tier as a Preform operation when it prepares for later steps. This marker does not hide, delete or disable the cut.

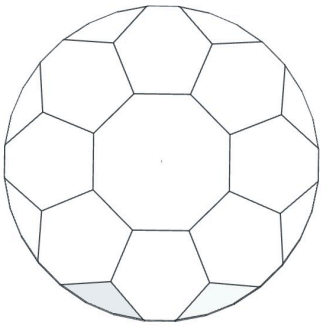
Name it for your next collaborator

Click a tier name to rename it, for example “Crown mains” or “Four-way decoration”. Toggle visibility to inspect effects; drag non-table tiers to reorder. These changes may invalidate later Meet sources.

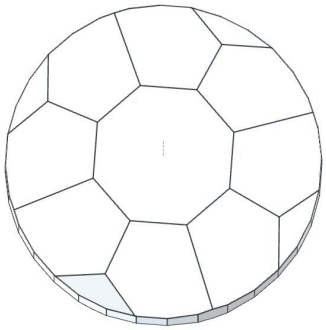
EXAMPLE E · CUSTOM RHYTHM

From eight repeats to four main directions

Goal: concentrate decoration in four directions to establish hierarchy. Custom indices specify where facets appear; the primary facet specifies the direction used for positioning.



Eight directions · 57 facets



Four directions · 53 facets

Variant 06 retains only I6, 30, 54 and 78. Removing four decorative facets changes the open areas; it does not remove all crown facets.

1. **Specify the positions**
   Import 02-edge-third.json, edit C2, switch to Custom indices and enter 6, 30, 54, 78.
2. **Choose the primary facet**
   Select 6 as the Primary facet index, then check the A Meet. The primary facet must belong to the set; it cannot be removed while a constraint uses it.
3. **Compare the composition**
   Use top view to check whether the four directions suit your intended setting. Recheck the solid after rotating or changing the index set.

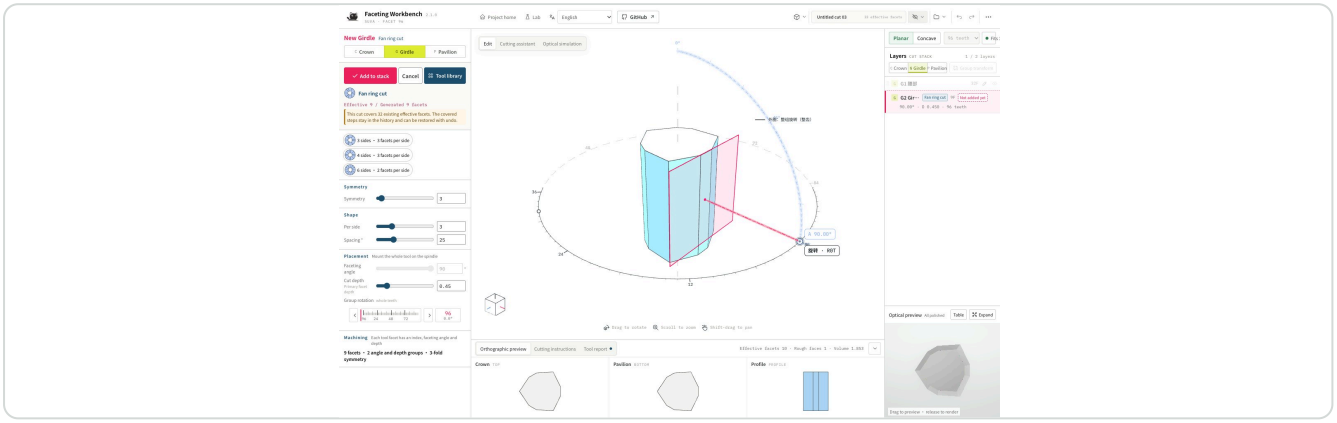
Before you continue

This example still has fourfold symmetry. Asymmetric sets are supported, but require facet-by-facet inspection; symmetry is not added automatically.

Ring cut · A symmetric group of facets

Round a triangle with a ring cut

To replace each side of a triangle, square or similar outline with a fan of facets, you no longer type indices one by one. A ring cut computes the whole group from symmetry and facets per side, snapped to the current wheel.



96-tooth default start, girdle ring L3×3, spacing 25°, cut depth 0.450: all 9 facets preview together and the top view turns the cylinder into a rounded triangle. Actual spacing 26.25°, offsets ±7 teeth per side.

1. **Start from the same point**
   Create a default 96-tooth project, choose Girdle on the left, start a cut layer and select Fan ring cut in the tool library.

2. **Set the outline and subdivisions**
   Set symmetry 3 and 3 subdivisions per side. Change spacing from 15° to 25°. Compare the tool hood with the lower crown top view to see how fan facets round the triangular corners.

3. **Set the depth and add it**
   Set cut depth to 0.450, check that all 9 facets are effective, then Add to stack. The index tape rotates the whole group by whole teeth; on the crown and pavilion you also adjust the angle as usual.

4. **Keep editing or dissolve**
   Click Edit on the layer to change symmetry, subdivisions and spacing again. When you no longer need the ring parameters, click Dissolve in the list; the facets stay as an ordinary layer and you can undo it.

Pause and make a design judgment

From 15° to 25°, are the triangle's corners noticeably rounder? Do you want to keep three clear sides or an almost round outline?

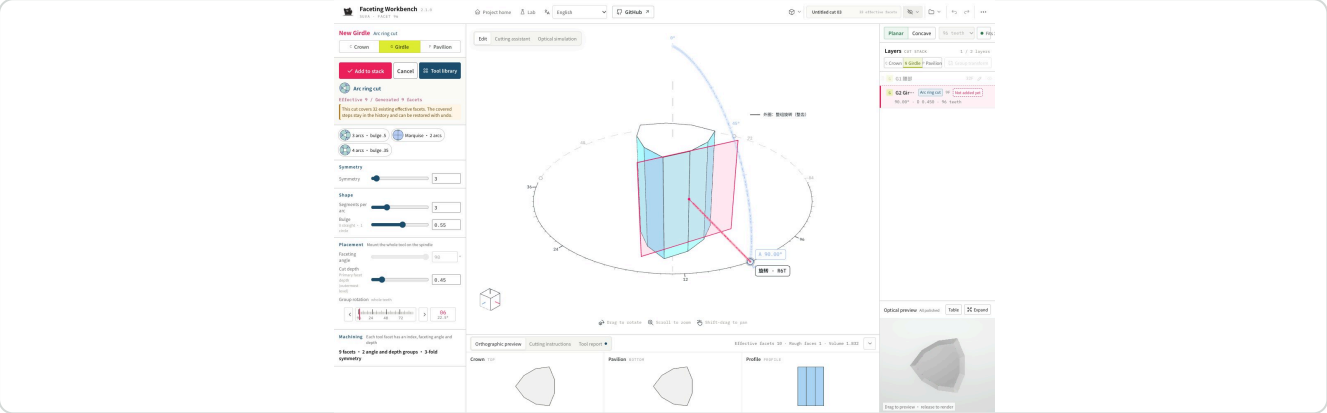
Before you continue

When the wheel cannot divide the symmetry evenly (such as 5 sides on 96 teeth), side centres round to the nearest tooth, so side spacing differs by 1 tooth and the composer says so. A fan ring cut shares one angle and depth; for depths that follow an arc, use the arc cut on the next page.

Ring cut · Arc · linked depths

Bulge a triangle into arcs with an arc cut

A fan ring cut uses one depth for the whole group. To bulge each side into a true arc, with the middle facets cut deeper, use the arc form: set the bulge, and the arc is split into equal chords whose depths are solved together along the arc, while you adjust only one depth.



96-tooth default start, girdle arc cut L3, 3 segments per arc, bulge 0.55, cut depth 0.450: all 9 facets preview together; the summary shows 2 depth levels, distance from centre 100%×6 · 92.5%×3, and the middle facet of each side cuts deeper than its neighbours.

1. **Start from the same point**
   Create a default 96-tooth project. Choose Girdle on the left, start a cut layer and select Arc ring cut in the tool library.

2. **Set the outline and bulge**
   Set symmetry 3 and 3 segments per arc. Drag bulge from 0 to 1, then settle on 0.55. The canvas shows the tool hood; Tool report lists depth levels and Orthographic views show how far the sides bulge.

3. **Set the depth and add it**
   Cut depth 0.450 sets the primary facet (the outermost level); the other facets cut deeper by their distance from the centre. Check that all 9 facets are effective, then Add to stack; the list shows Arc L3×3 · 2 levels.

4. **Dissolve to adjust level by level**
   To meet or re-depth one level on its own, click Dissolve on the layer and confirm Split into 2 layers by depth: every facet keeps its depth, now as two ordinary layers G2 and G2b, and you can undo it.

Pause and make a design judgment

From bulge 0 to 1, do the triangle's sides bulge as much as you want? How much deeper should the middle level be for the outline you have in mind?

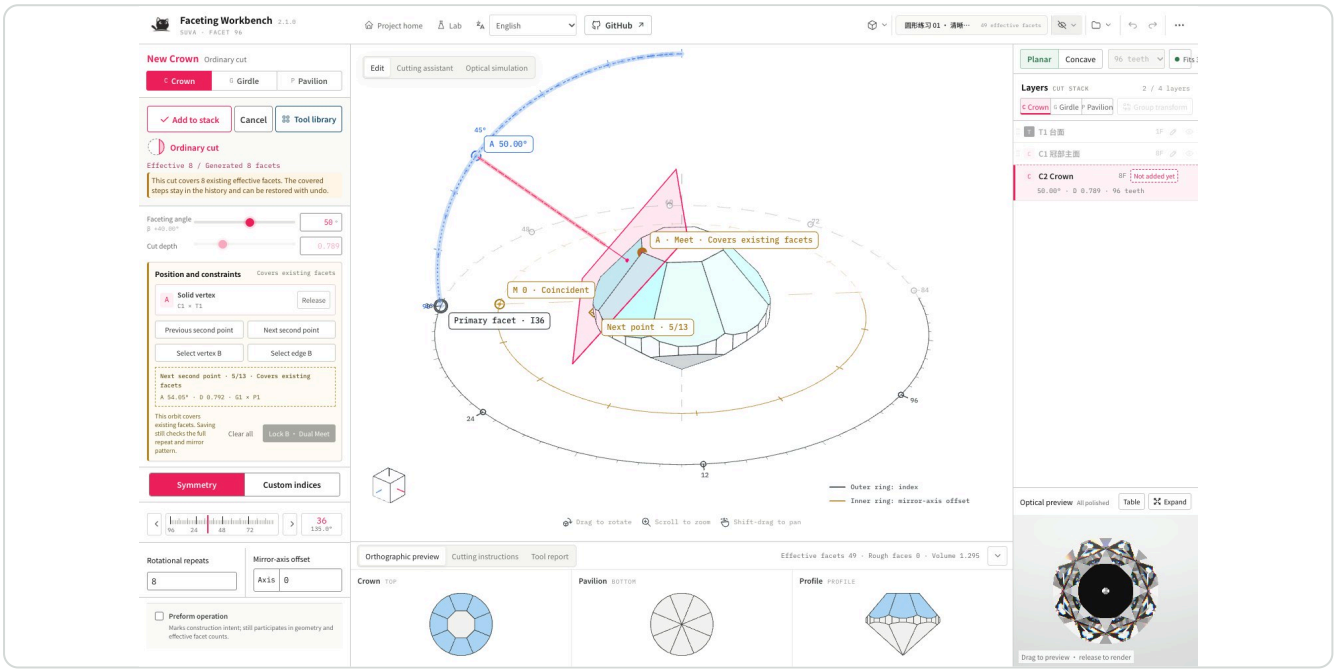
Before you continue

Symmetry can be 2 (at least 2 segments per arc and bulge 0.05) for a marquise-like lens. Per-facet Meet needs a dissolve first; ASC / GCS exports write one tier per depth, and the arc parameters are kept only in JSON.

EXAMPLE B · JUMP AND SINGLE MEET

Pass the primary plane through an existing intersection

Goal: meet an existing structure precisely. Jump browses candidate positions; Meet remembers a chosen position as a constraint.



After locking A, the target point determines depth; faceting angle remains adjustable.

1. **Find a position**
   Create a crown cut and use Next / Previous intersection, or Select vertex and pick a point on the solid. Check that the highlighted sources are the intended intersection.

2. **Lock A**
   Click Lock Meet A after inspecting the preview. When you change angle, depth adjusts to keep the primary plane passing through A.

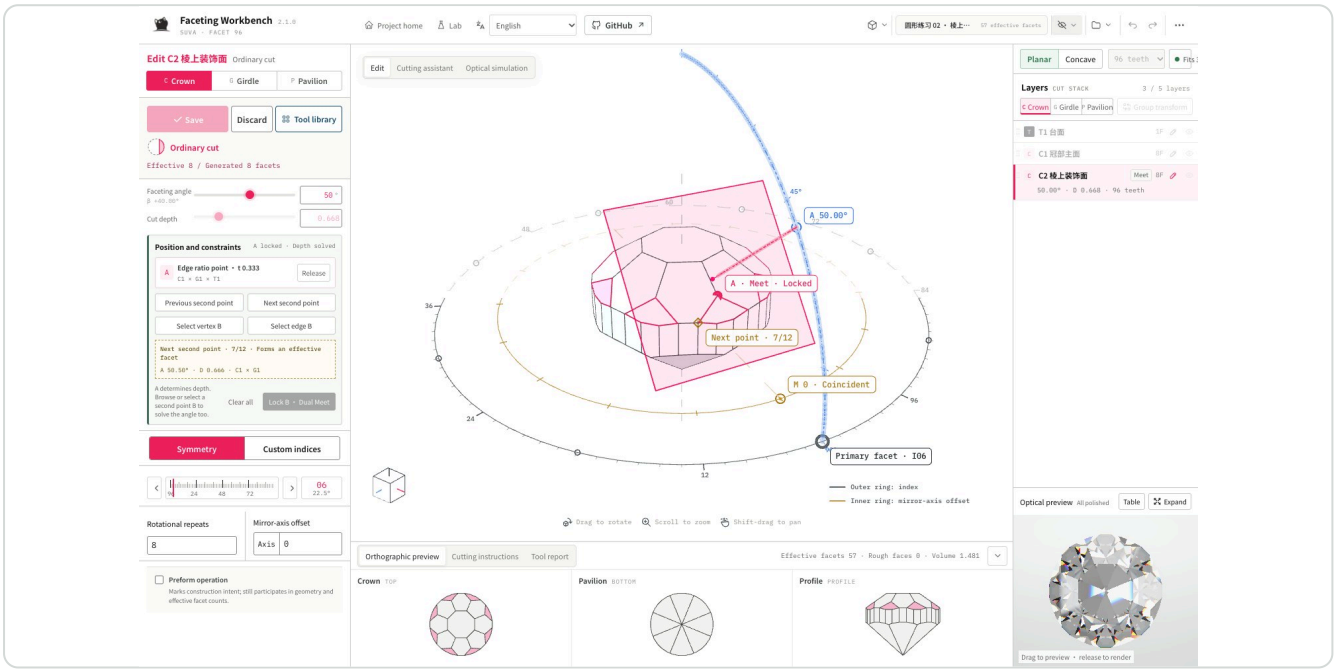
3. **Check the whole ring**
   Repeated facets cut together, but the constraint applies to the primary facet. Rotate the stone and check other directions before adding the cut.

> **Before you continue**
> This example adds a single-point crown cut to the round exercise. Later edge examples use the same starting point. Girdle and fixed table cuts do not support Meet / Jump.

EXAMPLE C · EDGE RATIO

Choose where decoration starts on an existing edge

Goal: add small facets between eight crown mains, turning sharp ridges into a decorative band of varying width. An edge ratio gives you a repeatable position to compare.



Variant 02: primary I6, 8 repeats, 50°, passing through one third of the sloping edge between C1's I96 and I12 facets.

1. **Identify the edge**
   Import 02-edge-third.json and edit C2. The A source description and canvas marker identify the sloping edge, not a horizontal table edge.

2. **Try from the beginning**
   Import 01, add a crown cut, and set index 6, repeat 8 and angle 50°. Select the sloping edge, set its ratio to 1/3, then lock A.

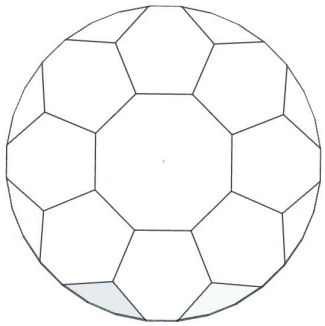
3. **Check the target position**
   Here the stable edge direction runs from girdle toward table, so 1/3 is nearer the girdle. For another edge, temporarily try t = 0 and 1 to identify its ends. Do not assume every edge runs upward.

> **Before you continue**
> An edge Meet passes the plane through one point on an edge; it does not place the whole edge in the plane. If you chose the wrong edge, select the correct source instead of compensating with depth.

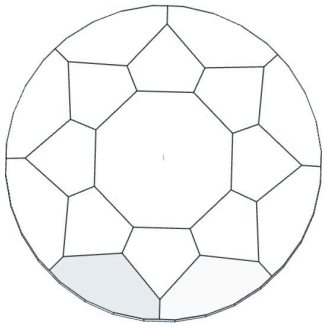
EXAMPLE C · COMPARE SHAPES

One edge, two ratios, two rhythms

Keep 50°, I6 and 8 repeats unchanged; move A from 1/3 to 2/3 of the same edge. This isolates how much crown area the decoration occupies.



1/3 · Variant 02



2/3 · Variant 03

Both files retain 57 effective facets. Equal facet counts do not imply equal facet widths or equal space around the table.

Workflow

Import 02, edit C2, release A, then select the same sloping edge at 2/3. Lock Meet A and save. You can also import 03 directly. A locked Meet is not adjusted by freely dragging depth.

Observe three relationships

Look at the new facet tips near the table, the remaining crown-main area and the eightfold rhythm in top view. Check axonometric view to rule out a viewpoint illusion.

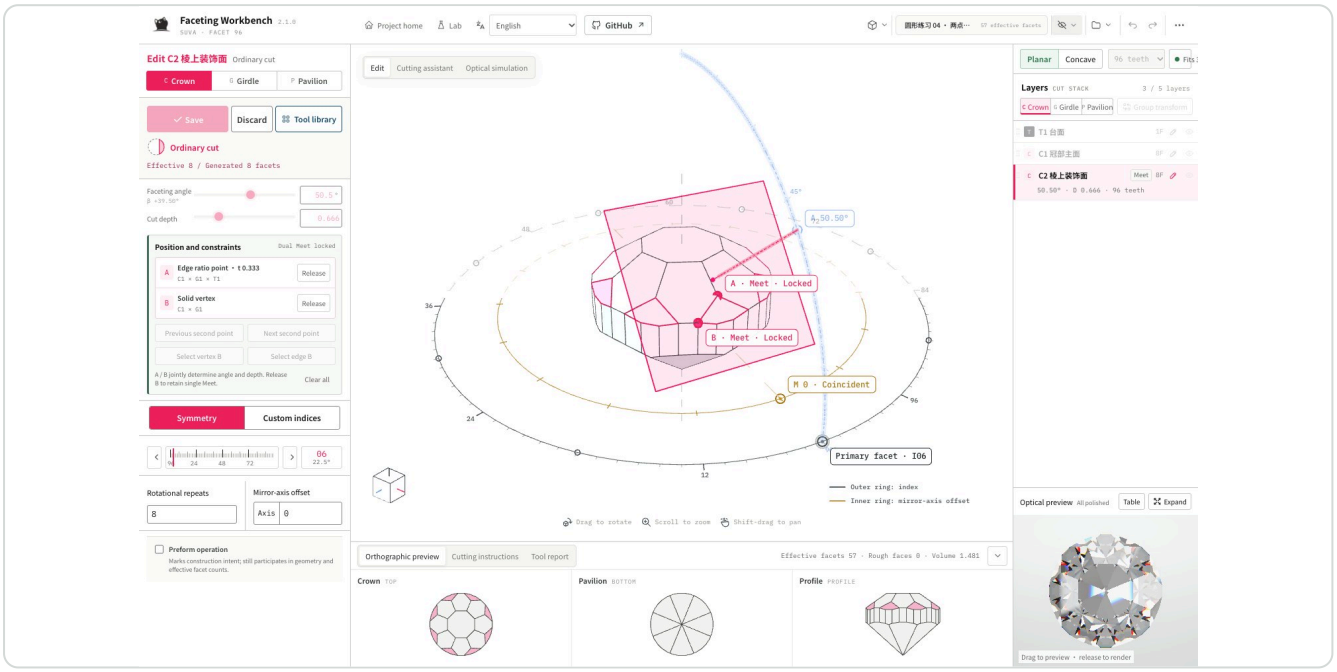
Let the designer decide

Do you want broader crown mains or more subdivision? Write a reason to use each alternative before choosing. This manual does not declare one more beautiful.

EXAMPLE D · DUAL MEET

Use a second point to determine tilt

Goal: satisfy two meeting relationships at once. With a fixed primary index, two distinct points can determine the plane's tilt and position together.



Variant 04: A is at one third of the sloping edge, B an existing vertex. The solved angle is about 50.50°; depth is solved with it.

1. **Inspect A and B**
   Import 04-dual-meet.json and edit C2. Inspect both markers and sources to understand which intersections they preserve.

2. **Try a second point**
   Starting with A locked, browse Next / Previous second point or select a second vertex or edge point. Inspect the candidate, then click Lock B · Dual Meet.

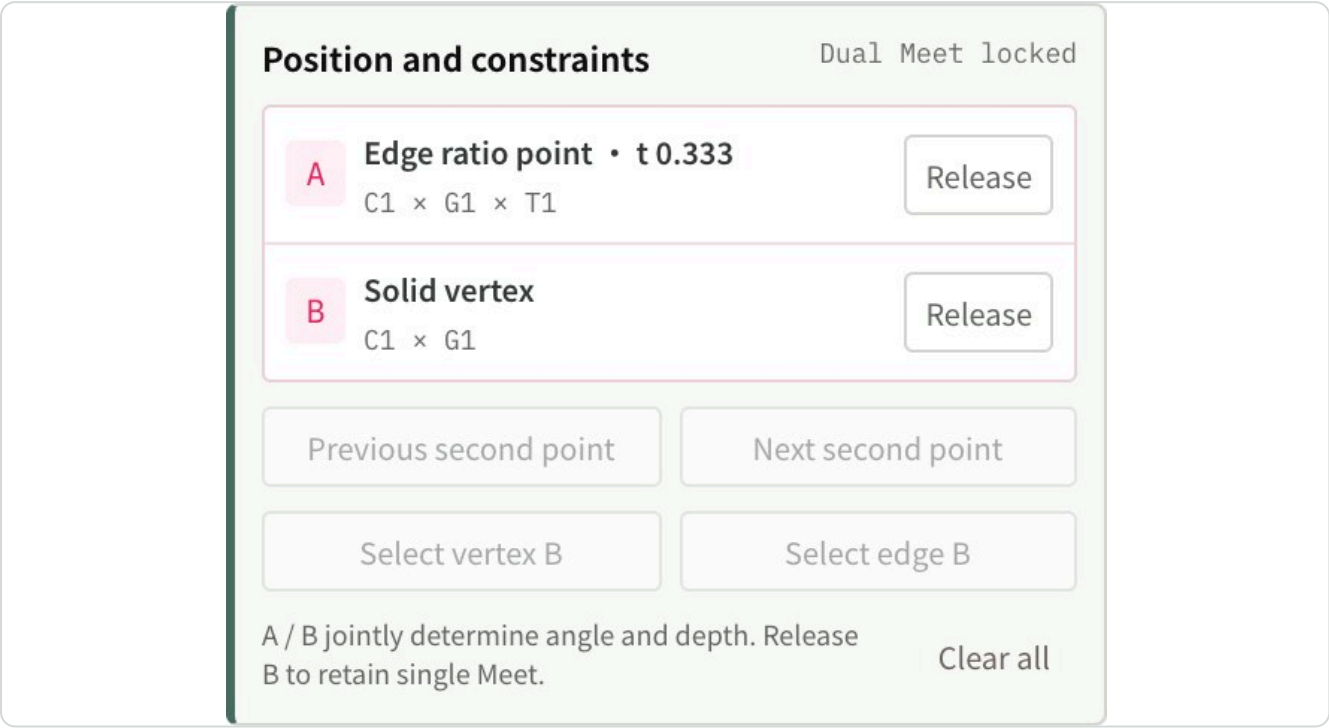
3. **Inspect the result**
   Angle and depth becoming read-only is expected. Use top and axonometric views to judge whether meeting both points is worth changing your preferred tilt.

> **Before you continue**
> A suitable second-point solution may not exist. Selecting two points does not make a design better by itself; dual Meet is a precision construction tool.

MEET · CANCEL AND RELEASE

Step back while keeping the choice you made

When exploring, know how to return. Distinguish browsing a second point from having both points locked.



A and B are shown separately so either source can be released. Locked parameters are visibly disabled.

B is still a preview

Canceling the second-point preview restores the original single-A angle and depth. You do not need to remember or re-enter them.

B is locked

Release B to retain A and adjust angle again. Release A to promote B to A. Clear all to restore free angle and depth.

When no solution is available

If points coincide, the direction is unsuitable or the target is out of reach, change the point or primary index. The system does not silently choose another primary facet.

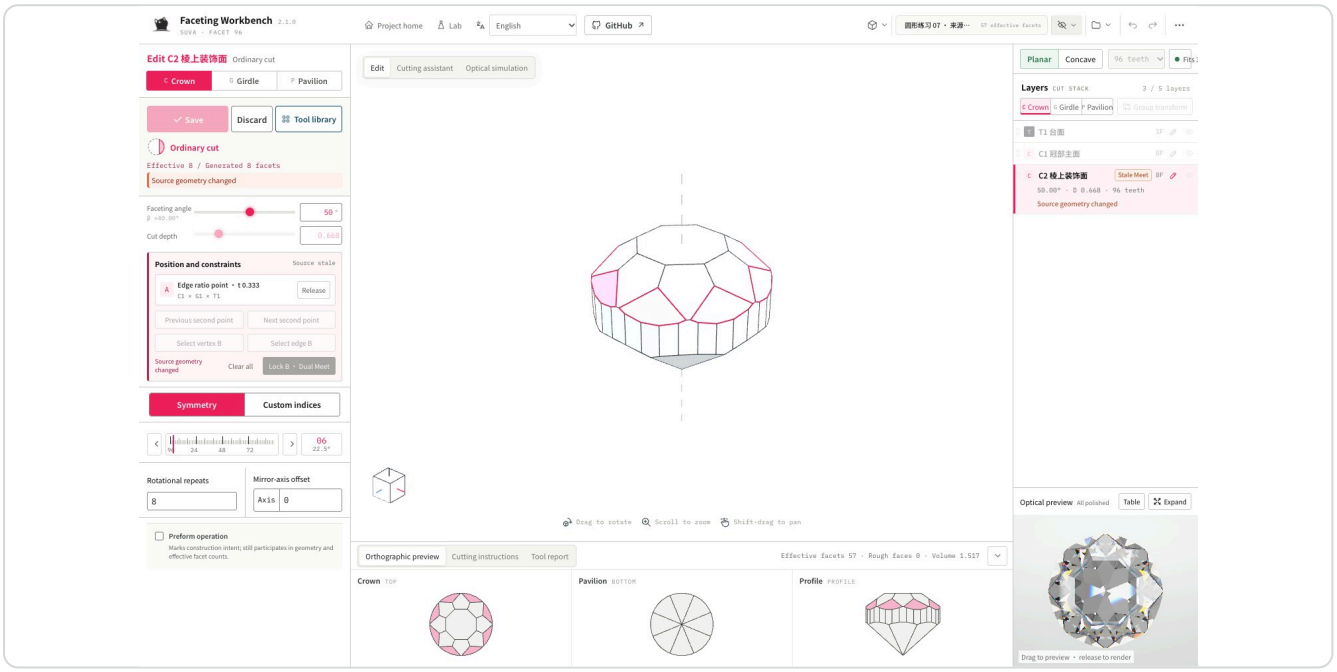
Escape order

Close the floating panel or current picking / ratio tool first, then leave the second-point preview, then cancel the CUT. Watch the state rather than pressing repeatedly.

EXAMPLE F · REPAIR REFERENCES

After changing an earlier shape, what happens to Meets?

Goal: retain saved cutting planes while identifying meeting relationships that need a new design decision. A stale source does not automatically recut the stone; the warning asks you to reconsider intent.



Variant 07: raising C1 Crown mains by 0.03 changes the source of C2’s original edge point.

1. **Read the warning**
   Import 07-source-changed.json and inspect C2’s stale marker. Compare it with 02: which intersections no longer match your intent?

2. **Select again**
   Edit C2, release the stale source, and select the required edge point or vertex on the solid before this cut. To retain the original intent, try one third of the same sloping edge.

3. **Decide again**
   Inspect and save the new preview. If the Meet is no longer needed, clear the constraint and retain a free construction. Do not choose an arbitrary point just to remove the warning.

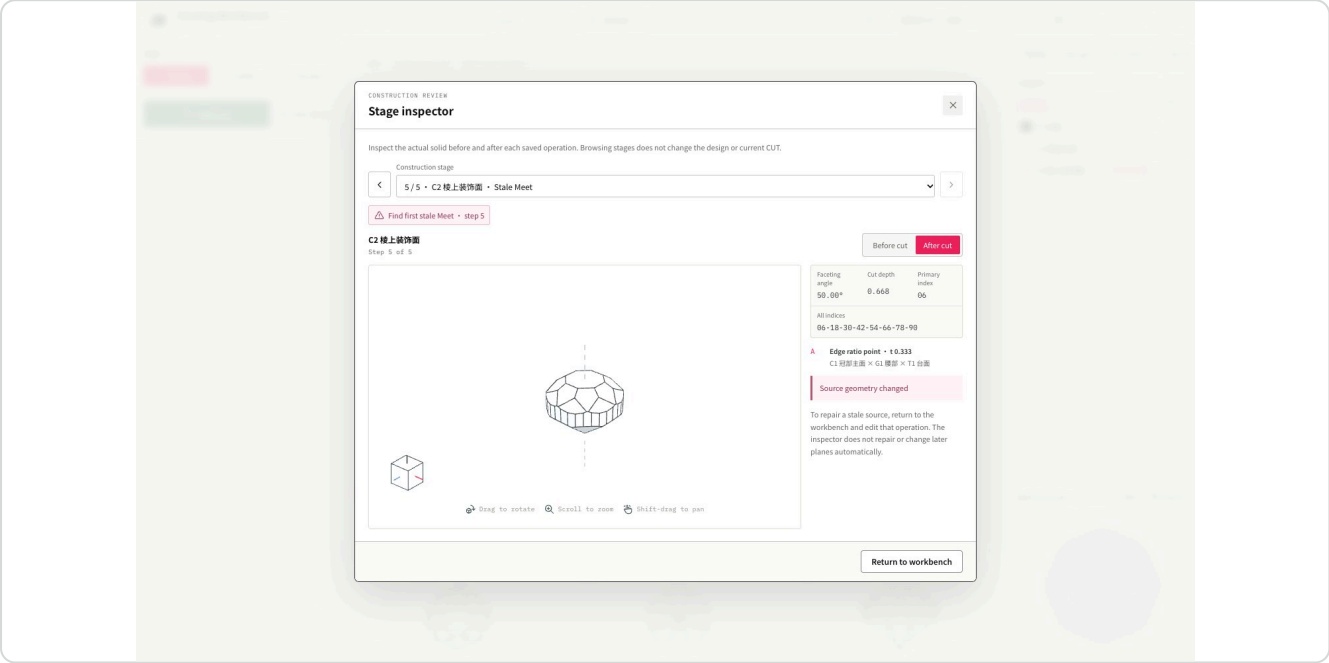
Before you continue

Hiding, deleting, reordering or editing a source can make it stale. Saved planes remain intact and a warning appears; downstream cuts are not all changed automatically.

DIAGNOSE · STAGE INSPECTOR

Inspect stages to understand how the shape changed

Goal: break the shape into understandable design steps. Use the inspector to review mains, decoration and source relationships, alone or with another designer.



The inspector pairs stage geometry with parameters and sources. This example focuses on C2’s stale reference.

1. **Open**
   More tools → Stage inspector. Select an operation or locate the first stale Meet.

2. **Compare before and after**
   Switch Before cut / After cut to see what the step actually leaves. A hidden operation produces no difference.

3. **Return with a question**
   The inspector is read-only and preserves your editing workspace. Close it and edit the relevant tier to repair the design.

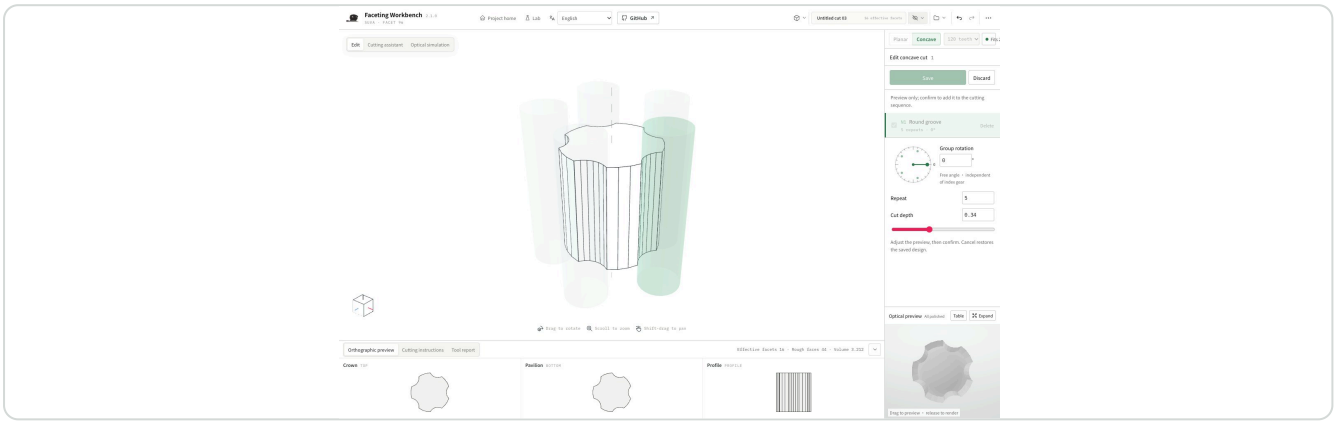
> **Pause and make a design judgment**
> Does this step add clear structure or unnecessary fragments? Does it still serve as a preform? Keeping or reverting it are both valid choices.

> **Before you continue**
> To review meets across the whole stone, use More tools → Meetpoint inspection. It lists candidates such as nearby meet points, short edges, small facets and girdle height differences for diagnosis only and never blocks saving; with concave cuts enabled it says they are not supported yet.

CONCAVE LAYERS · REPEAT, ROTATE & DEPTH

Add five grooves with one cutter

Choose a cutter and see how it changes the outline. No coordinates, axes or long parameter forms are needed.



The same 120-index default project, with a round groove added in the right Concave panel: five repeats, fixed-frame depth 0.34.

1 **Add a cutter**

Use the 120-tooth project from page 14. Switch to Concave, then New concave cut → Round groove. The translucent cutter and five grooves are a preview; click Add to stack to save this starting point.

2 **Adjust the two controls**

Click the saved round groove and change depth from 0.34 to 0.50. Watch the main and orthographic views. Releasing the pointer keeps a preview; Save commits it, while Discard restores 0.34.

3 **Compare repetition and shape**

Edit again, change repeats from 5 to 6 and rotation to 17.5°. Compare groove positions from the same view, then Save these changes together. Outside editing you can enable or disable a layer; planar-view visibility offers a temporary comparison.

4 **Restore your preferred design**

Undo the two saves to return to five repeats, 0° and depth 0.34. Each explicit save is one undo step, regardless of slider movements. Switch to Planar and toggle concave visibility to compare outlines without changing saved data. Triangular cutters also offer tip angle, width and length while retaining cutting depth.

Pause and make a design judgment

Do deeper or more numerous grooves preserve the girdle width and edges you want? Does undo restore the original fivefold shape?

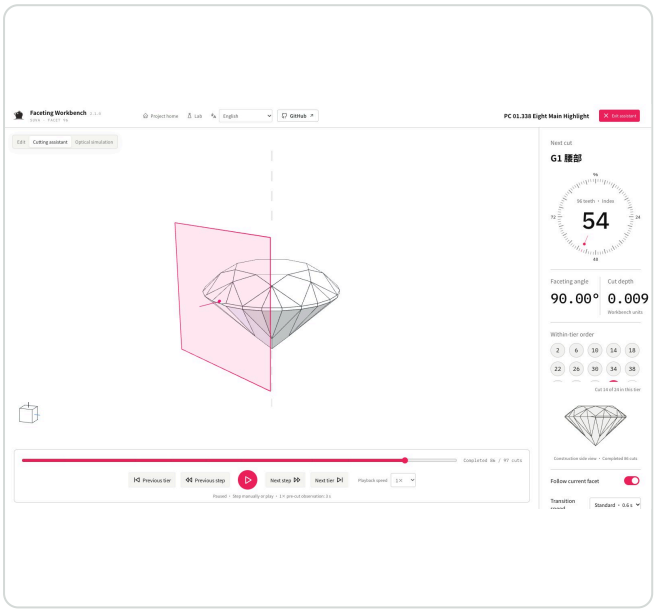
Before you continue

V groove uses an ideal 90° edged wheel. A thick blank may retain its top and bottom edges. Changes can be undone; the simulation does not guarantee machining or polishing results.

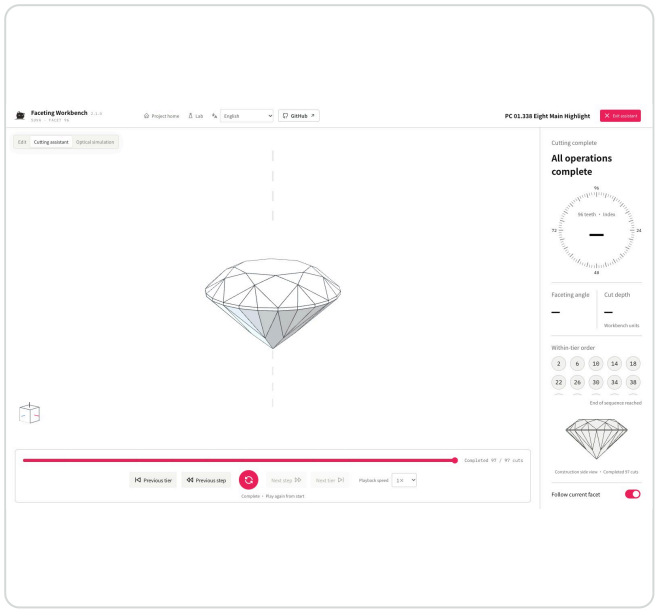
DEMONSTRATE · CUTTING ASSISTANT

Replay the actual cutting order before delivery

Check the intended cutting order or show a collaborator how the stone is built. The assistant replays the committed design from the rough, one cut at a time, in CUT STACK order.



86/97 cuts completed · Next cut and intermediate shape



Step 97/97 · Complete

Actual replay of Eight Main Highlight: the pink plane marks the next planar cut; the partial solid shows completed steps. The sequence uses committed planar cuts and enabled concave layers, excluding unsaved drafts.

1. **Enter**
   Choose Cutting assistant in the upper-left mode switch. Step 0 shows untouched stock. Hidden planar layers are skipped; the fixed table is one step. All planar cuts run first, followed by enabled concave layers. Each concave layer includes all its repeated tools in one clearly labeled whole-group step.

2. **Watch by cut or by tier**
   Use Next / Previous step on the bottom bar for individual cuts, or Next / Previous tier to jump groups. Play runs at 0.5× / 1× / 2×. Drag the timeline to seek; manual steps and backgrounding pause playback. The right panel shows the next index, angle, depth and within-tier order. Completed counts cuts already made. Frosted facets show “Frosted facet · Do not polish” without changing the cut count or parameters.

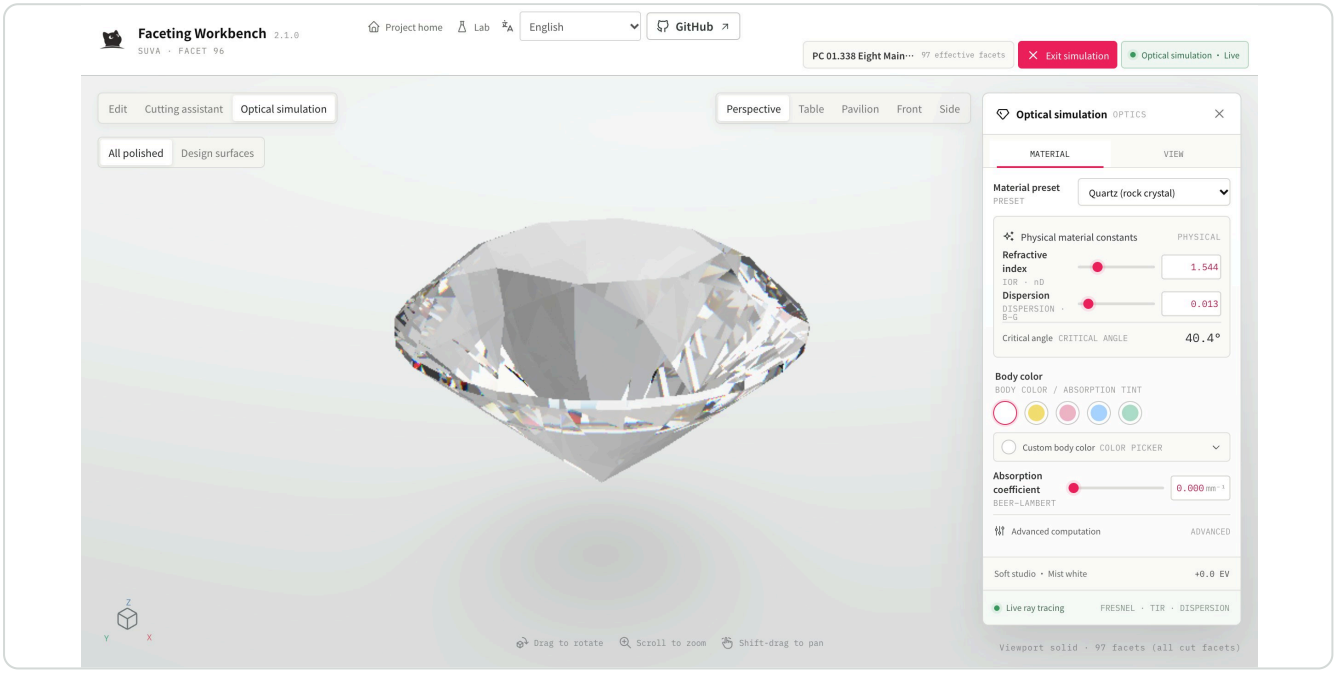
3. **Return to editing**
   Click the pink Exit assistant button or press Escape to restore the previous workspace. The assistant is read-only and adds no history. After each cut, it holds 0.5 seconds, then eases to a 45° offset view; the next plane appears once settled. Adjust transition speed on the right. Dragging pauses following; the switch resumes it. Use Stage inspector for stale-source diagnosis.

> **Pause and make a design judgment**
> Which facet group’s ordering needs explanation to the faceter? Does the sequence express your intent, or should you revise CUT STACK?

OPTICS · OBSERVE DIFFERENCES

Fix viewing conditions before comparing materials

The quick optical preview starts face-up. Drag or scroll independently; double-click or choose Table to reset. Geometry updates during interaction; optics render on release. Further input cancels the old task. Expand opens the full simulation while preserving cutting drafts.



Use the upper-right bar to switch views. This actual quartz-material simulation of Eight Main Highlight shows one viewing condition, not a prediction of physical appearance.

Keep conditions consistent

Use the same material, body color, studio, exposure and viewpoint when comparing two cuts, so differences remain attributable.

Compare frosted and polished finishes

The preview opens in All polished. Select Design surfaces at the upper left to preview frosted finishes saved in the laboratory; unmarked faces remain polished. Keep the material, environment and camera unchanged, then let the image settle before comparing. Switching modes does not change finish annotations, cutting parameters or undo history.

Save each alternative

Material is saved with the document. Keep separate JSON files for A/B alternatives and record viewing conditions. Exiting simulation does not commit unsaved cuts.

Before you continue

Simulation supports design comparison; it does not guarantee physical fire, brightness scores or yield. Material, size, cutting accuracy and viewing environment also affect the real result.

OPTICS AND GEOMETRY · DESIGN JUDGMENT

Four questions for a useful design review

Do not choose only the prettiest perspective image. Examine both alternatives in the same order and explain why you keep or reject each.

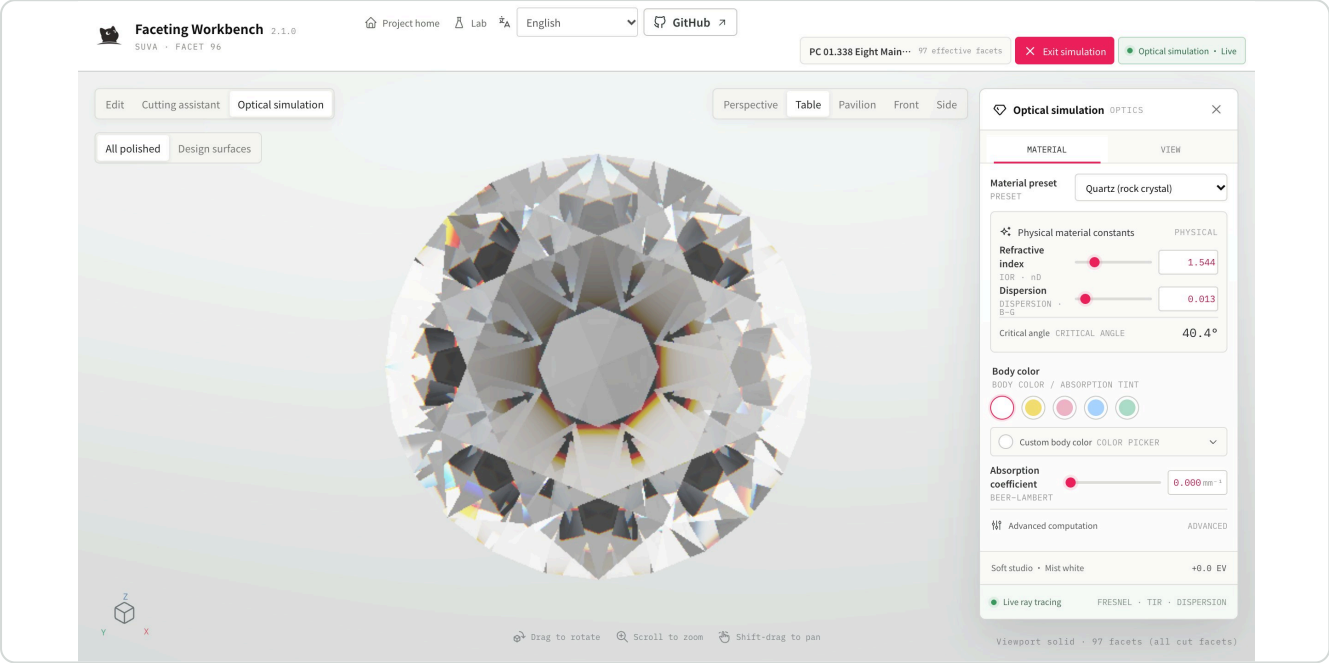


Table view makes outlines and facet groups easier to compare. Rotate slightly to understand changes in light distribution.

> **Top: does the rhythm work?**
> Do main and decorative facet proportions support the goal? Is the space around the table intentional or overcrowded?

> **Front and side: do the proportions fit?**
> Does the combination of crown height, girdle and pavilion depth suit the intended form and setting direction?

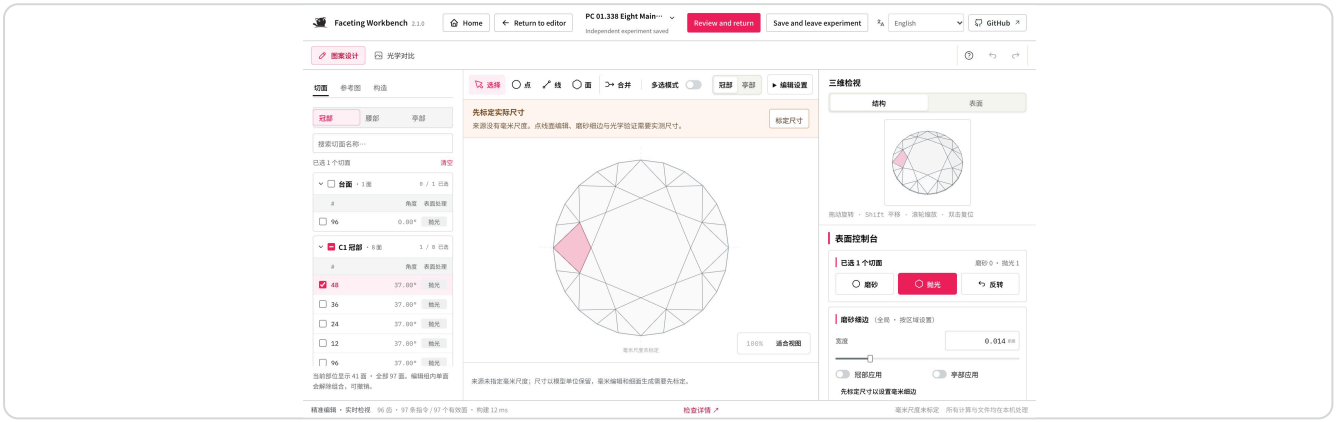
> **Rotate: can you explain the changes?**
> With fixed material and exposure, rotate slightly and compare light structure. Avoid hiding large dark areas with overexposure or mistaking studio reflections for facet edges.

> **Return to geometry: is the shape clear?**
> Exit simulation and check intersections, missing facets and remaining rough. An optical image does not replace a geometry check.

NAVIGATE · LAB

Bring an experimental design back as a new version

Open the pattern laboratory from home or the editor to explore surface patterns. Drafts save independently and return as new projects. Create new crown/pavilion shapes in the editor tool library; older preset-lab drafts remain available.



The workspace has one brand header. Select the experiment name to inspect its source version or download recovery data.

1. **Choose a lab and bring in a design**

   Save or discard the current cut, then open the pattern laboratory. Bring in the current design with its source snapshot, or create a new experiment with a gear and millimeter width.

2. **Edit and review**

   Select facets by layer or with multi-select, then choose frosted, polished or invert in the surface console. Frosted borders are off by default; enable crown and pavilion separately as needed, using a shared width. Choose Check and return, review operations, effective facets, scale and source status, then Return as a new project. If the source changed, the experiment still returns from its original snapshot.

3. **Resume an older crown/pavilion draft**

   Open an existing crown/pavilion draft from Recent experiments; its components and source remain intact. Save and exit or inspect and return. For new designs, use Tier assemblies in the editor tool library.

4. **Exit and recover**

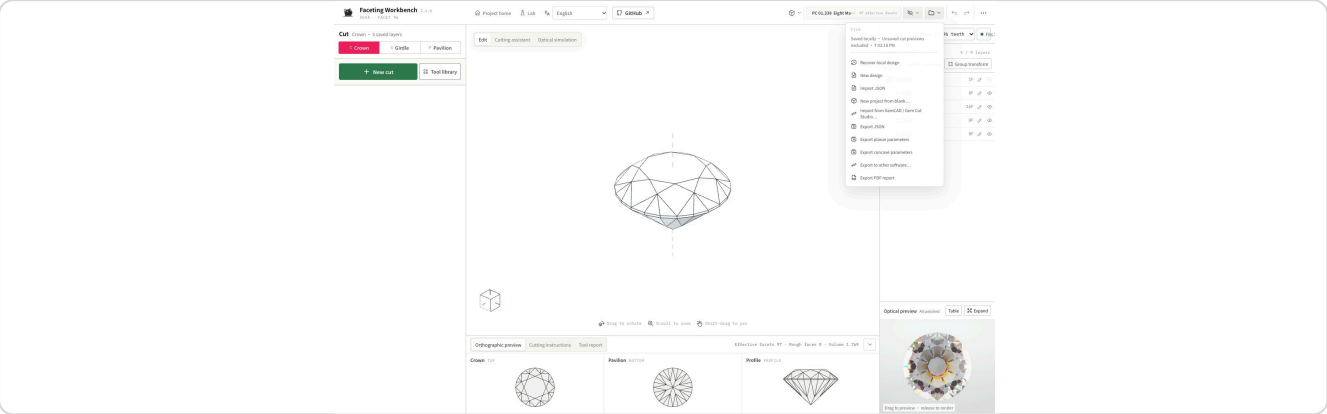
   Resume saved work from Recent experiments. On a conflict, download a recovery file and use Save from recovery file in the overview. Returned frosted facets appear in cool gray; the assistant marks them as needing no polish.

> **Before you continue**
>
> Set a measured width with Calibrate size before dragging points or adding frosted borders. Invalid old recipes retain their actual planes. The pattern lab accepts planar composite-tool facets, but not mesh rough or concave cuts; older preset-lab drafts can retain bounded convex mesh. Lab interfaces are currently Chinese.

FILES · SAVE AND DELIVER

Deliver the design and the ability to keep editing

Provide an editable master file, a report for discussion and a clear design note, alongside any presentation image.



Floating menus close on an outside click or Esc. The File menu offers JSON, PDF and Export to other software…, which opens the format center. Save the CUT first so the export is the design you mean to deliver.

Continue designing: JSON

Preserves stock, planar cuts, concave tools, each tier's wheel, Meet sources, preform intent and optical material. Keep this as the master file. Unsaved previews are excluded.

Read and discuss: PDF

The technical report draws the complete committed solid and lists planar parameters on the selected wheel. Concave tools have a separate appendix. Choose whether to include girdle rows. When the design has frosted faces, it exports as all polished by default, or can mark frosted faces in the views and facet tables. Then check the name, tooth count, units and equipment requirements.

Hand to other software: ASC and GCS

File → Export to other software… opens the format center, where you compare what GemCAD text and Gem Cut Studio keep, simplify or lose. Only final effective planes are exchanged; enabled concave cuts or an imported mesh stock block the export, and the full operations stay in JSON.

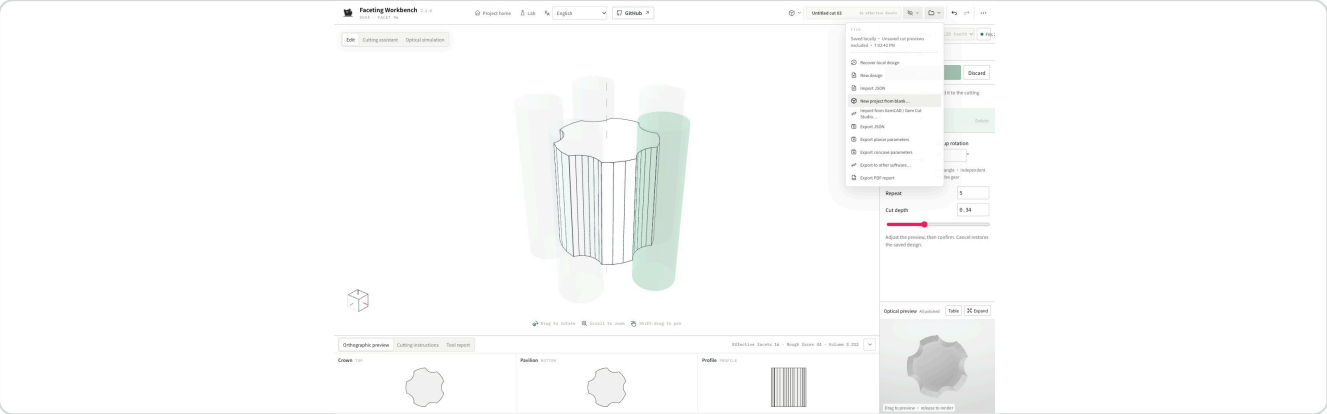
Before you continue

To take over designs from GemCAD or Gem Cut Studio, see page 34. Wheels such as 120 or 360 and fractional directions are kept, not forced to 96.

Saving designs · Reusing parameters

Save and export from the File menu

All exports are in the top File menu. The left sidebar holds planar parameters; the blank is fixed at creation. Switch between planar and concave layers on the right.



The File menu offers JSON, planar parameters, concave parameters, PDF, and import/export through the format center; a solid stock also allows exporting stock parameters.

1. **Save the full design**
   In the five-groove project, open File and export JSON. It preserves the wheel, stock, planar cuts and concave cutters.

2. **Check the wheel**
   Before exporting JSON or PDF, check the selected and compatible wheels. Never read 120-wheel indices directly on a 96 wheel; red means whole teeth do not match.

3. **Choose the appropriate format**
   Stock and planar parameters are in the same menu; PDF is for reading the report. A complete design with enabled concave cuts cannot convert to ASC or GCS, and the format center explains why.

4. **Check the restored shape**
   Import the complete JSON as a new project and compare top and side views. The wheel, planar cuts and five grooves should match; the original project remains.

Pause and make a design judgment

Can you find every export in one place? Does the report wheel match the equipment you intend to use?

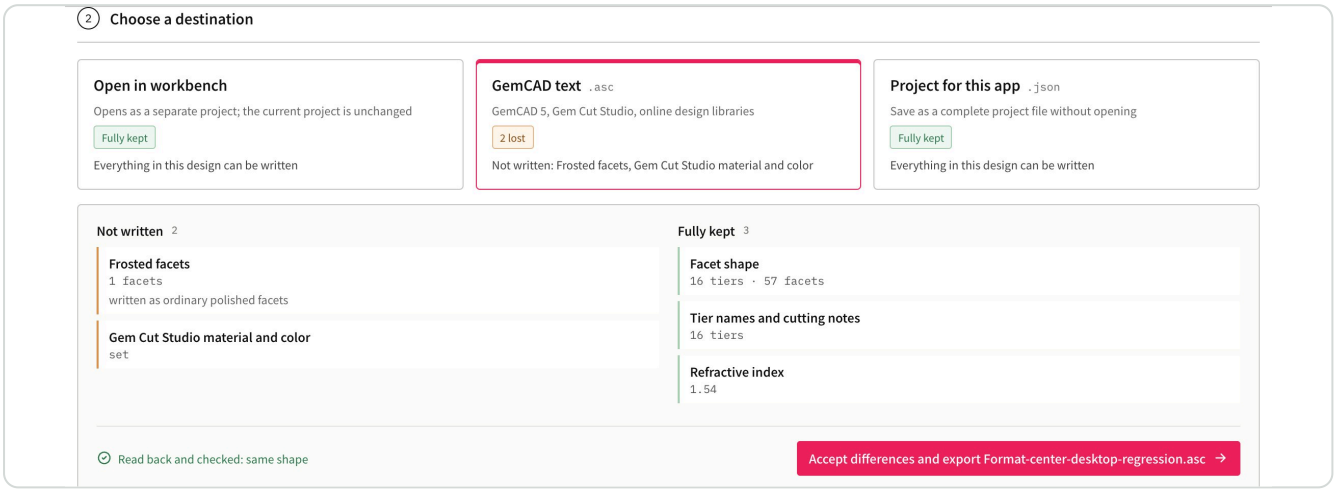
Before you continue

Keep complete JSON for important projects: planar parameters are required, stock and round tools are optional. Unspecified indexing means 96; compatibility is recalculated from actual directions.

Format center · See the differences before converting

Take over GemCAD and Gem Cut Studio designs

Old designs need not be redrawn. The format center reads GemCAD (.asc, .gem) and Gem Cut Studio (.gcs) and can hand your design back to them, listing what is kept and what is lost before every conversion.



A square emerald saved by Gem Cut Studio 1.1 (with one frosted face), with GemCAD text selected: the card and report state that frosted facets and the Gem Cut Studio material are not written, everything else is kept, and the exported file was read back and checked.

1. **Open the format center**
   Project home → Format center; or in the editor, File → Import from GemCAD / Gem Cut Studio….
2. **Choose a source**
   Drop an .asc, .gem or .gcs file and check the preview, wheel, tier / facet counts and frosted facets. Reading never changes the original file or the current project.
3. **Compare destinations**
   Select Open in workbench, GemCAD text and Project for this app in turn, and read each card's result and the Not written / Simplified / Fully kept lists below.
4. **Decide and continue**
   Open in workbench creates a separate project. When an export has differences the button reads Accept differences and export, and one click completes it; the export is read back to check the shape.

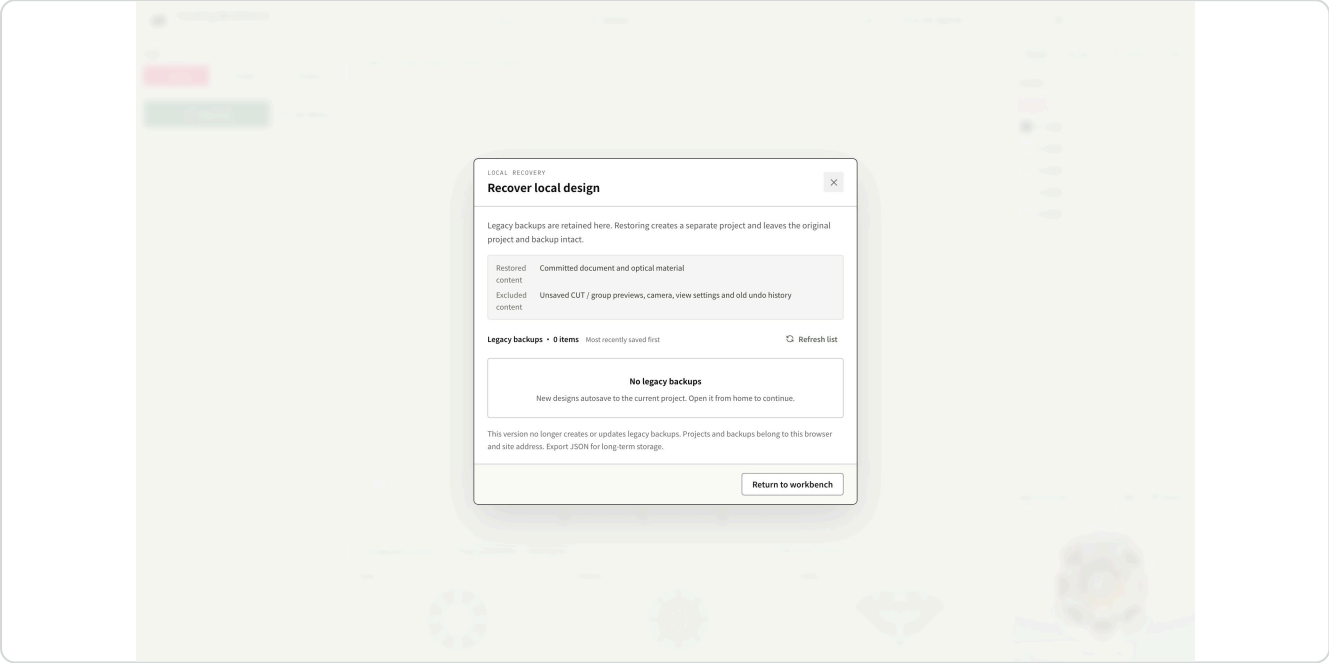
> **Pause and make a design judgment**
> Does the information this conversion drops matter for what you are handing over? Keep a JSON as the complete master if you will keep editing.

> **Before you continue**
> GemCAD .gem is read only. Designs without a single table convert directly between ASC and GCS but cannot open in the workbench. Gem Cut Studio index directions are calibrated on real version 1.1 saves and were checked in Gem Cut Studio with a twisted design.

PROJECTS · RECOVERY · SHORTCUTS

Save projects and recover older designs

Projects retain the latest committed document and material but do not replace exported JSON archives. Different site addresses or browsers have separate project storage.



File → Recover local design lists legacy backups only. If empty, continue an autosaved project from home.

Project saving and legacy backups

Committed cuts and material save to the current project. Legacy backups migrate once to home and are no longer updated. Manual recovery creates a separate project and preserves both the original project and the backup. Drafts, camera and old history do not survive reload. Concurrent windows detect save conflicts and pause autosave; reload, save as a new project or export JSON to resolve them.

Useful shortcuts

Drag or use arrow keys to turn the near side of the gem in the input direction, consistently in Edit and Optical simulation. Shift-drag pans, scroll zooms, 0 resets. J / Shift+J browse candidates, M locks A, B locks a second point, V toggles vertex picking. Text fields prioritize typing.

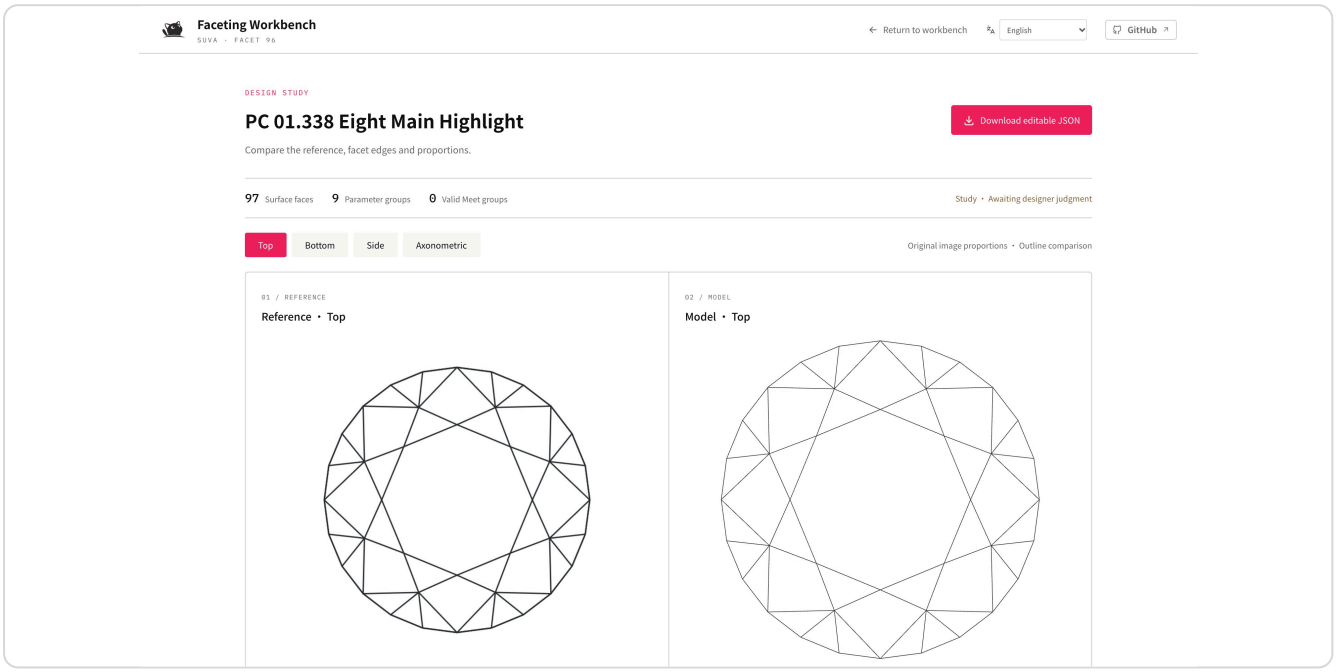
When you need an explanation

More tools → Help center. Escape closes help or the active menu without changing the design. If saving fails, retry, export JSON or use Manage local projects to delete unneeded projects and free space. Archive before deleting projects or clearing browser data.

CUT STUDY · CODEX ASSISTANCE

Turn a reference into an editable design

Bring a reference image and design intent. State the outline, table and girdle thickness to retain, then compare actual models and resolve unclear connections step by step.



Eight Main Highlight study page: the left reference is its technical top view; the right uses the same preset JSON. This demonstrates comparison workflow, not photographic reconstruction accuracy.

1. **Tell Codex your goal**
   Use facet-parametric-design in the workbench project with an image you have permission to use. Ask it to separate known details from inferences and deliver editable JSON with real-view comparisons. The workbench can edit and import independently without Codex.

2. **Compare under the same conditions**
   Open the generated study page and inspect top, bottom, side and axonometric views. Missing independent references are disclosed. Images must not be stretched to manufacture a match.

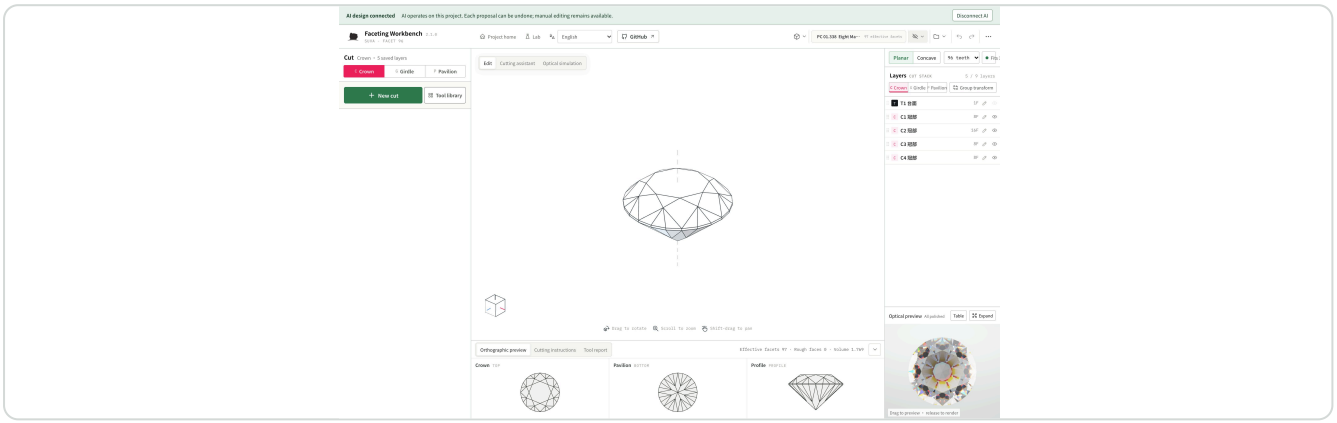
3. **Keep the study and refine it**
   Download editable JSON and import it through the file menu in a new project. Adjust one group’s angle or depth and inspect facet width and girdle thickness. Save the accepted variation and compare from the same angle.

> **Pause and make a design judgment**
> Which connection or proportion did this version resolve? Which areas still depend on inference and need a clearer reference or your judgment?

OPTIONAL LOCAL AI · CONVERSATIONAL DESIGN

Combine conversation and manual editing

Compare alternatives in conversation, then refine familiar parameters manually. Both use the same CUTs and undo history. The manual workbench requires neither Codex nor MCP.



Actual connected workbench: Eight Main Highlight, 97 effective facets. The following steps share one design between conversation and manual editing.

1. **Open the connected workbench**
   Ask an MCP-capable client to open the local workbench and follow its returned link. The top bar shows AI design connected. Ask the conversation to confirm the project name, then describe the outline to keep and the facets to change. Ordinary web use needs no connection.

2. **Compare before applying**
   Create and adjust all 18 tier and fancy tools in conversation: for example, change an eightfold crown from three steps to four. Preview before applying; undo if needed. Dissolve for individual edits. Save or discard manual edits before continuing.

3. **Refine manually and archive**
   Edit the target tier, adjust angle or depth, compare from the same view and save. Request editable JSON in conversation or export it from File. Use JSON as a reliable transfer between browsers or local ports.

4. **Return to manual design at any time**
   Click Disconnect AI at the top. The project, history and parameter controls remain usable. Users without AI can create, import, edit, save and export in the standalone static workbench.

Pause and make a design judgment

At the same top and side views, did the crown change retain the intended outline and meetings? Which steps suit conversational exploration, and which need your manual judgment?

Before you continue

This page checks collaboration workflow, not cutting or optical gains. See docs/mcp/README.md for MCP installation and development. Codex and other local MCP clients are supported.

DESIGN REVIEW · DELIVERY CHECKLIST

Complete a design, not just a sequence of clicks

Review the work as a designer: can you discover the controls, understand changes, make trade-offs and hand over the result? Successful interactions are the foundation, not the whole outcome.

State the goal

Use one sentence, for example: “Add lighter four-way decoration while retaining eight crown mains.”

Show before and after

Compare original and revised shapes side by side at the same angle. Identify the visible changes, beyond parameter screenshots.

Explain the choice

Describe your preferred outline, open areas or facet rhythm. Record remaining concerns and the next experiment.

Complete delivery

Ensure no CUT is unsaved; address stale sources; check multiple views; export the JSON master and PDF report, with ASC if needed.

Give actionable feedback

Describe “My goal → Where I clicked → What I saw → What I expected”, with a screenshot and JSON. Unclear controls and confusing meanings are valid feedback too.

Pause and make a design judgment

Can you explain when group proportions, edge ratios, dual Meet and a custom primary facet helped with your own design problem?

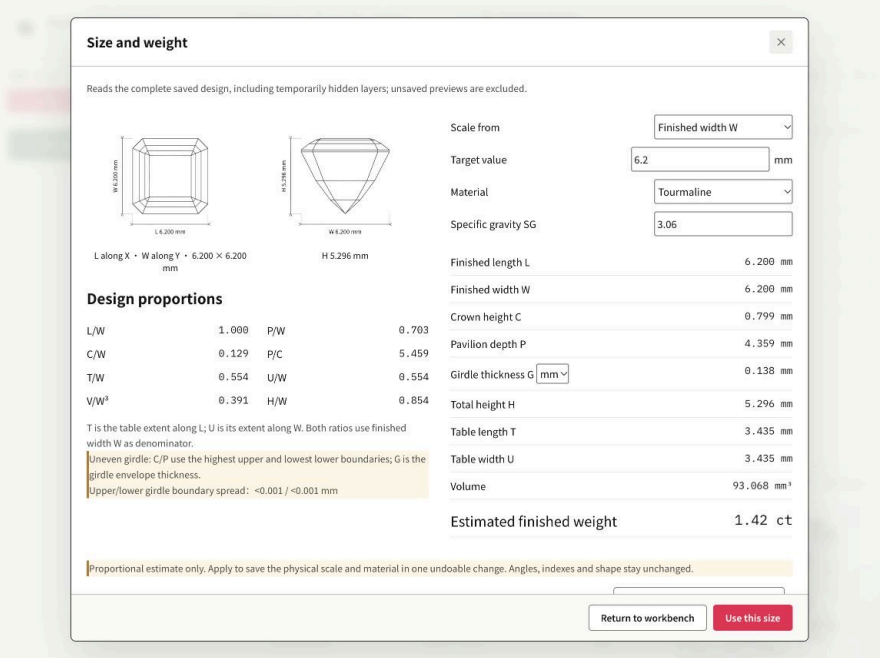
Before you continue

This manual and original exercises are provided by the faceting workbench. Preset authors, sources and third-party font / reference rights remain as documented. The applicable version is printed in the footer.

Size and weight · Delivery preparation

Determine finished dimensions and estimated weight

Create PC 07.001C Square Emerald 1:4 from the preset library. Choose the finished size, then estimate weight using material specific gravity. Scaling preserves angles, indices and geometry.



Real preset: width 6.20 mm, tourmaline SG 3.06, estimated weight about 1.424 ct. This is not rough yield.

1. **Find the size calculator**
   Open More tools → Size and weight; shortcuts also appear in cutting instructions and the cutting assistant. Choose finished width W and enter 6.20 mm.

2. **Compare size options**
   At 12.40 mm the estimated weight is about 11.391 ct, eight times the original. Trials do not change the saved design. Adopting a size is undoable.

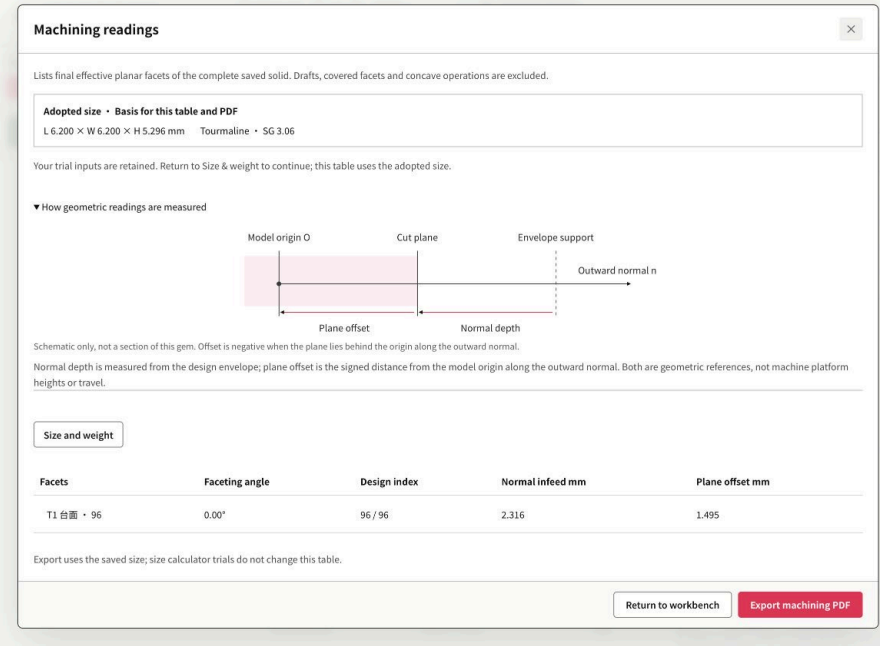
3. **Identify the material**
   Choose a preset SG, or select Custom and enter a material name and SG. A physical rough scale cannot be overwritten. Adjust estimated SG to measured data when available.

> **Pause and make a design judgment**
> At the same viewing angle, does the size change preserve your intended proportions? Does the delivery identify the size and material?

Geometric references · PDF

Hand geometric readings to the cutter

The machining table reads final effective planar facets from the complete saved design. Its header shows the adopted size and material. Calculator trials are retained when switching panels, but do not change this table.



Expand the reference diagram to distinguish origin-to-plane offset from inward depth measured from the envelope support plane. This is not a section of the current gem.

1. **Check the reference**
   Normal depth and plane offset are geometric references, not handpiece platform heights or travel. This feature does not adapt to a specific machine.

2. **Return to the trial or locate a facet**
   Use Size and weight to continue your previous trial. In the cutting assistant, select a facet and view it; exit the assistant before adopting a size.

3. **Export and verify**
   The machining PDF uses the adopted size. An English interface produces English explanations. Check indices, millimeter values and material names; the technical report also includes a size and weight page.

> **Pause and make a design judgment**
> Can another cutter identify the adopted size and reference from the file alone? If not, clarify them before treating any geometric value as a machine setting.